
MODEM for Radio Frequency Identification (RFID)

Oct. 7, 2005

Hanbat National University

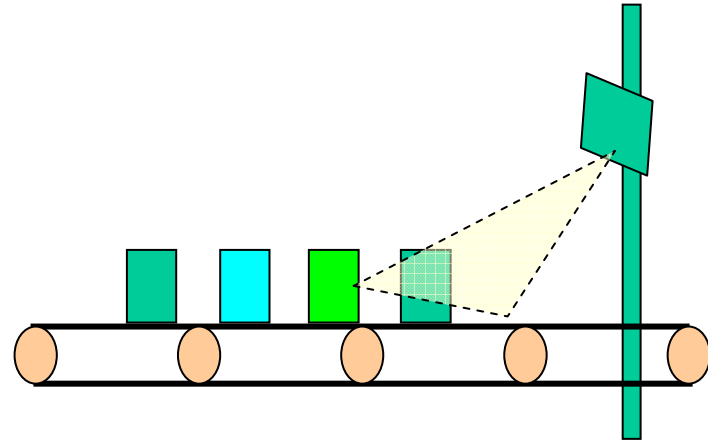
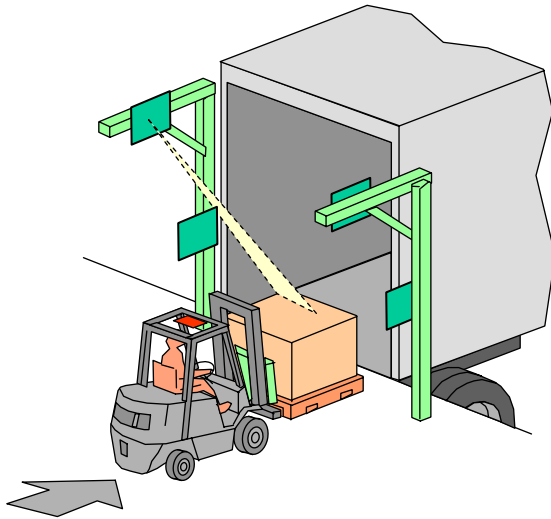
Hyung Chul Park

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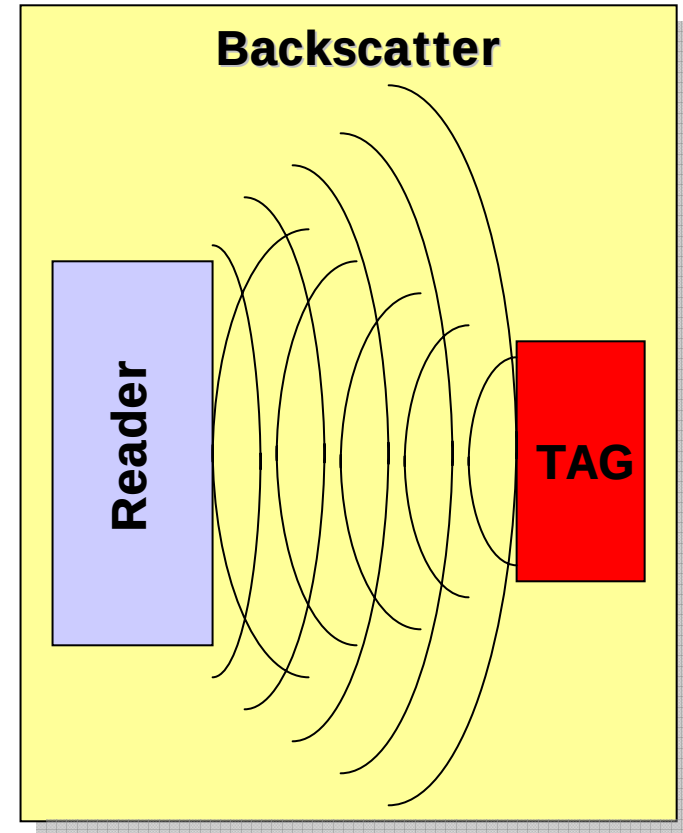
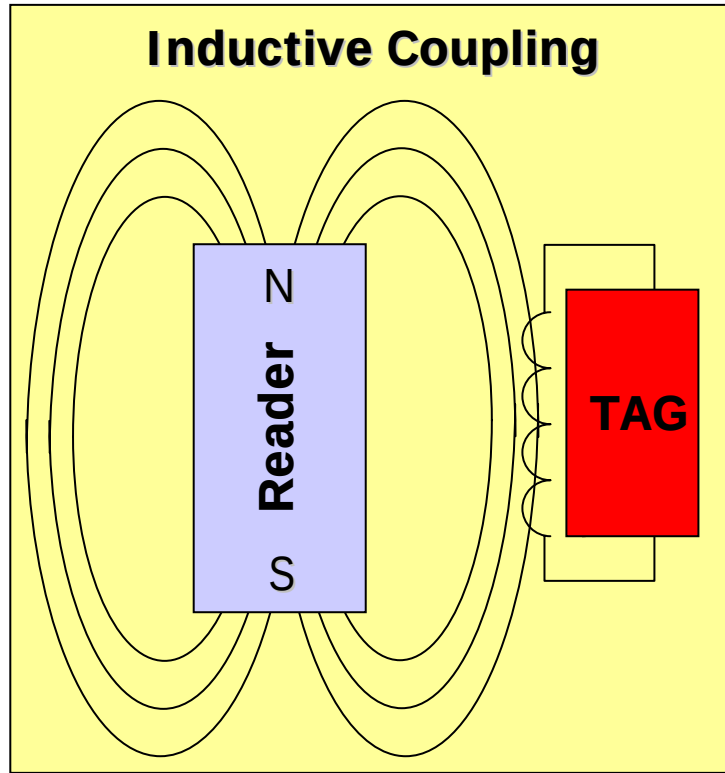
- What is RFID?
- EPCglobal RFID protocol (class-1, generation-2 UHF)
 - Features
 - Specifications for Interrogator → Tag transmission.
 - Specifications for Tag → Interrogator reception.
- Conclusions
- Agilent ADS system simulator for UHF RFID

Radio Frequency Identification

- Tag wirelessly sends bits of data when it is triggered by a reader.
- Superior capabilities to barcode:
 - Non Line of Sight.
 - Hi-speed, multiple reads.
 - Can read *and* write to tags.
 - Unit specific ID.



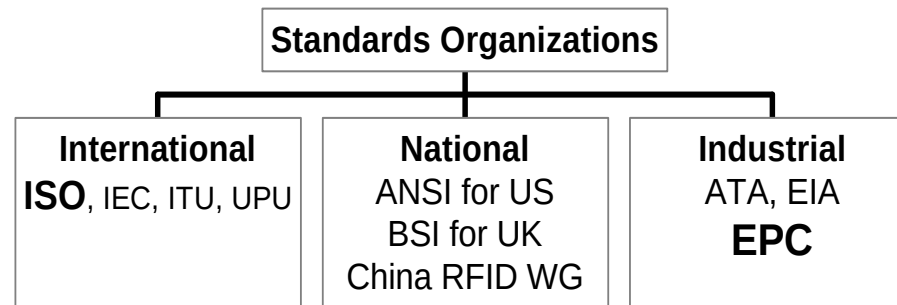
Basic Tag operational principles



- Near field (LF, HF): inductive coupling of tag to magnetic field circulating around antenna (like a transformer)
 - Varying magnetic flux induces current in tag.
 - Modulate tag load to communicate with reader.
- Far field (UHF, microwave): backscatter.
 - Modulate back scatter by changing antenna impedance.

EPCglobal RFID protocol:
Class-1, Generation-2 UHF RFID
at 860-960MHz

EPCglobal standard and standardization



Partnership between 100 global firms, including founders:

- Uniform Code Council
- EAN International
- Procter and Gamble
- Gillette



- Standards Development
- Adoption
- Brand management and marketing
- Policies (Privacy, Intellectual Property)



6 of them
Continued Research

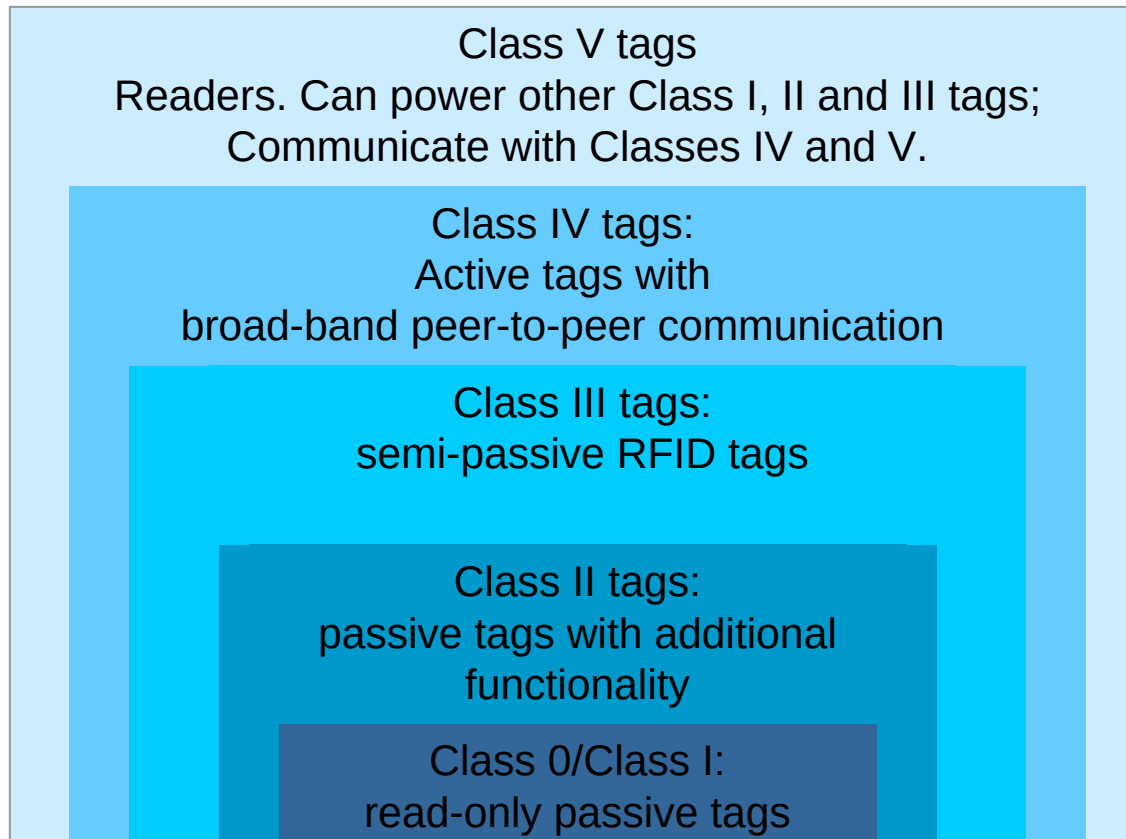


ISO 18000, part 6

Research → Commercialization

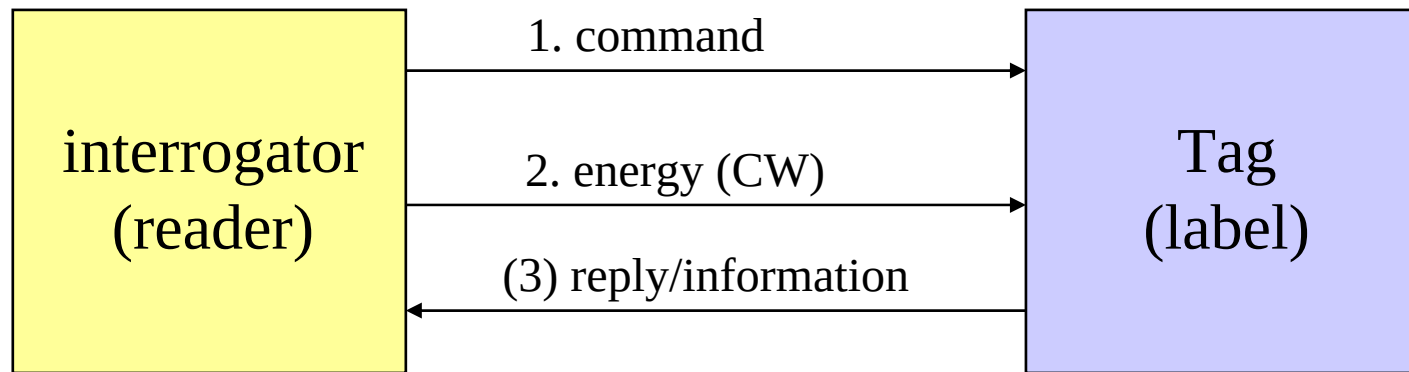
Features (1)

- Class-1 Tags
 - Identity Tags
 - Passive-backscatter Tags.



Features (2)

- Interrogator-talks-first (ITF).
- System
 - Interrogator (reader).
 - Tag (label).
- Half-duplex.
- Frequency range : 860~960MHz



EPCglobal RFID protocol:
Interrogator(Reader)→Tag transmission

Features (1)

- REMARK
 - All of the presented features can be modified for local regulations.
 - (R:~~~)
- Frequency range (R:range)
 - 860~960MHz
- Operating channels (R: no. of channels)
- Operating frequency accuracy (R: accuracy [tighter])
 - +/-10ppm : -25~+40°C
 - +/-20ppm : -40~+65°C
- Modulation
 - DSB-ASK, SSB-ASK, PR-ASK.
 - Double-sideband ASK
 - Single-sideband ASK
 - Phase-reversal ASK
- Data rates
 - 26.7kbps (data-1:2T_{ari})~128kbps (data-1:1.5T_{ari})

Features (2)

- Data encoding : PIE
 - Pulse-interval encoding
- Other features
 - Frequency hopping, channel bandwidth, etc.

Pulse interval encoding (PIE), Data rates

- The interval between two pulses determines the meaning.
 - data-0 : shorter duration of CW transmission.
 - data-1 : longer duration of CW transmission.

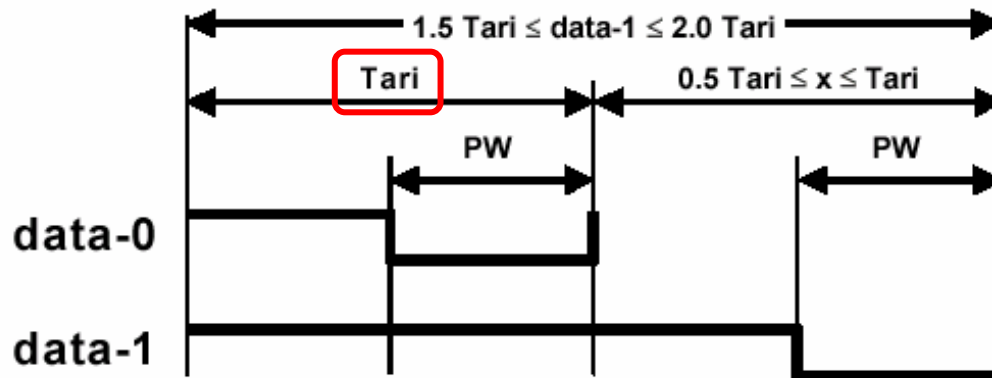


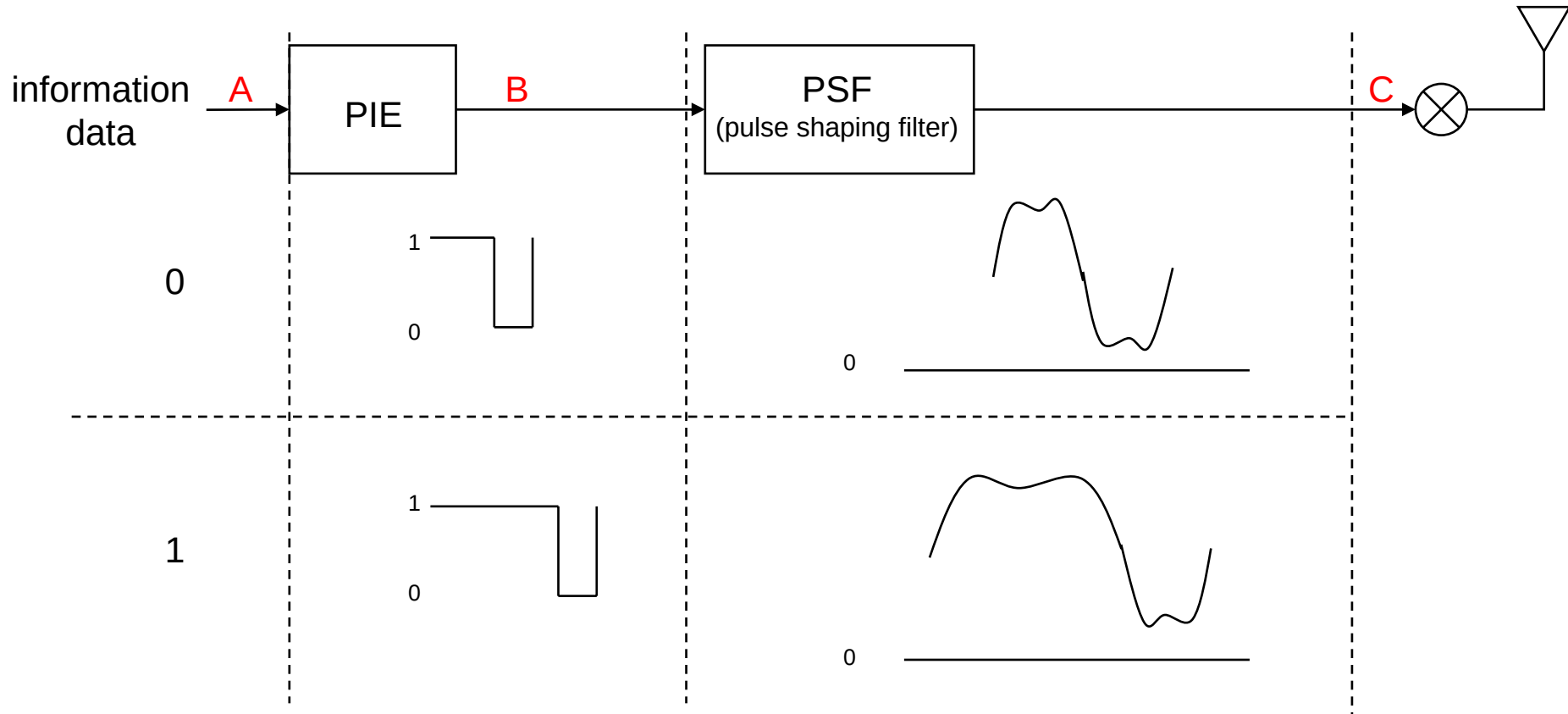
Figure 6.1 – PIE symbols

	T_{ari}	Tolerance of T_{ari}	Spectrum
128kbps (data-1: $1.5T_{ari}$)	6.25usec	+/- 1%	DSB-ASK, SSB-ASK, Or PR-ASK
	12.5usec		
26.7kbps (data-1: $2T_{ari}$)	25usec		

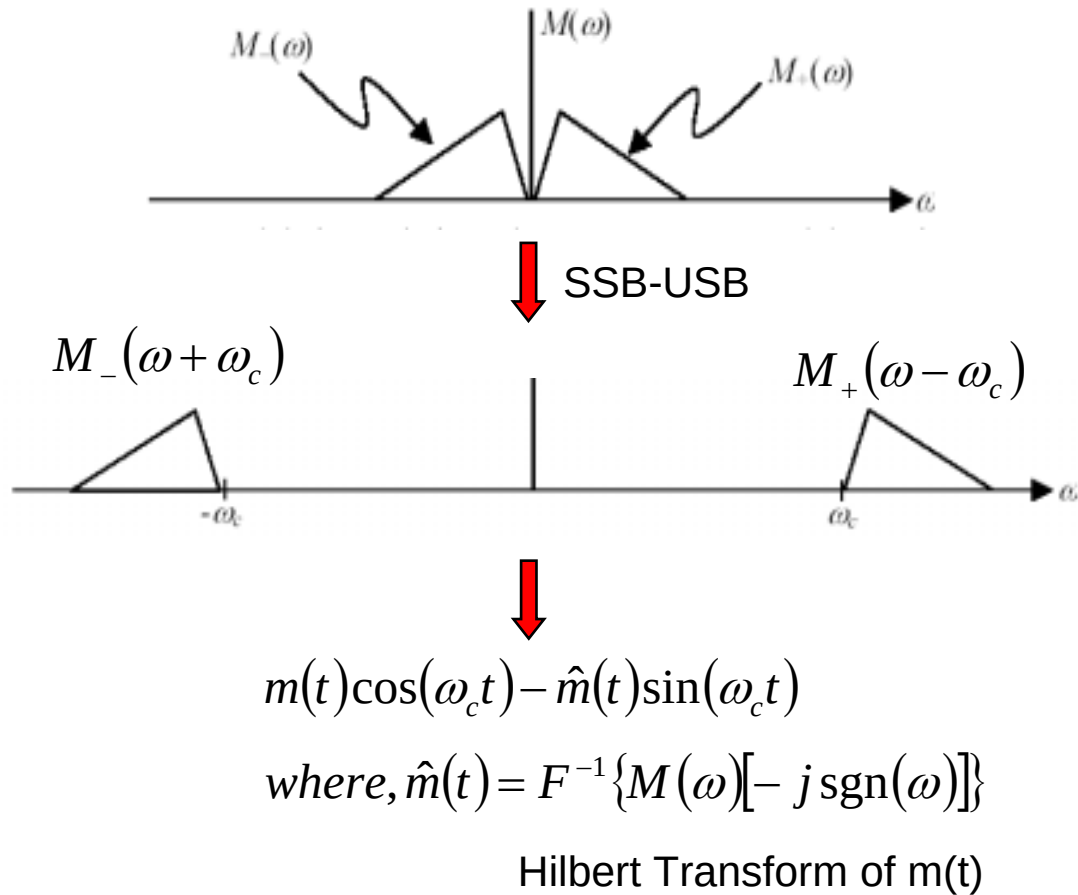
Modulations

- Single carrier modulation
 - Double side-band ASK (DSB-ASK)
 - Phase reversal ASK (PR-ASK)
- Quadrature carrier modulation
 - Single side-band ASK (SSB-ASK)

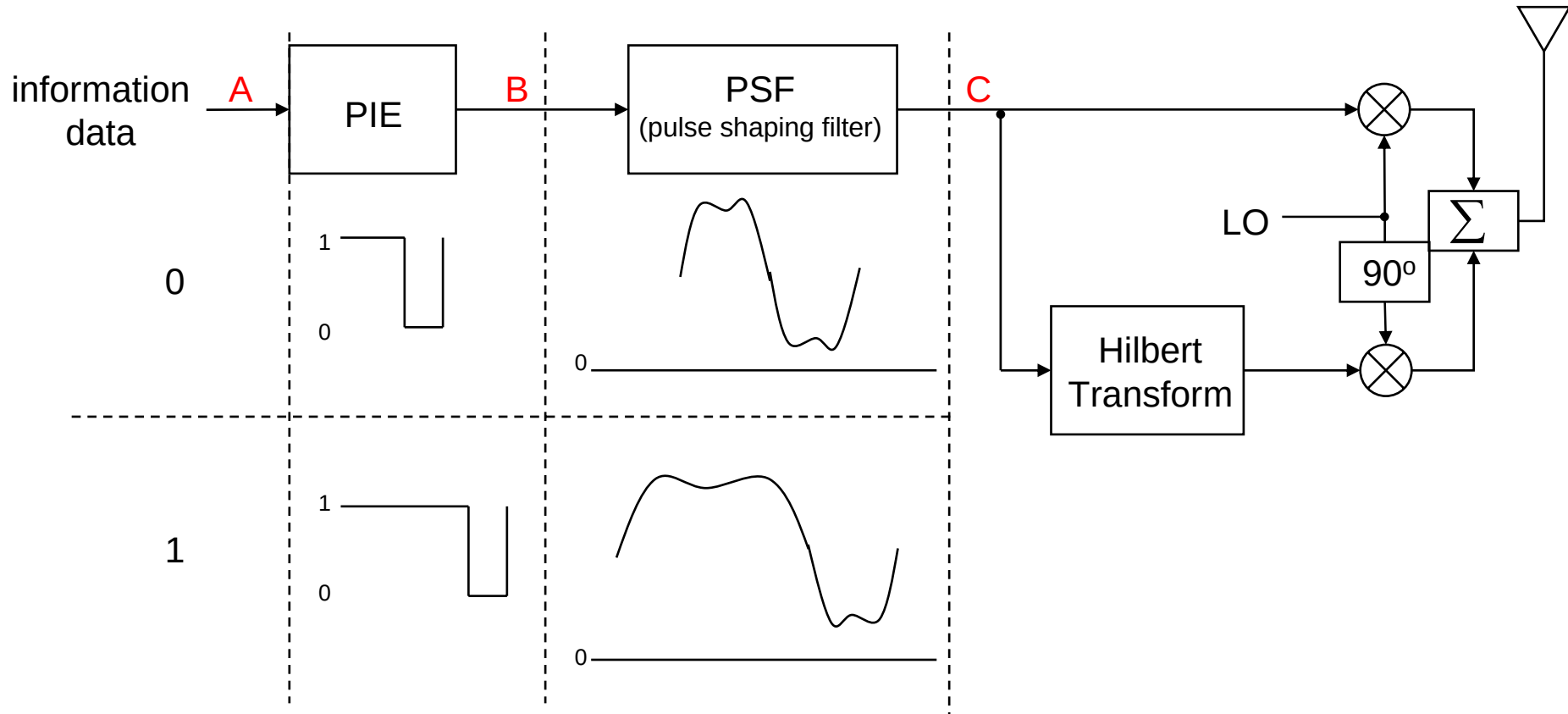
DSB-ASK



SSB-ASK (1)

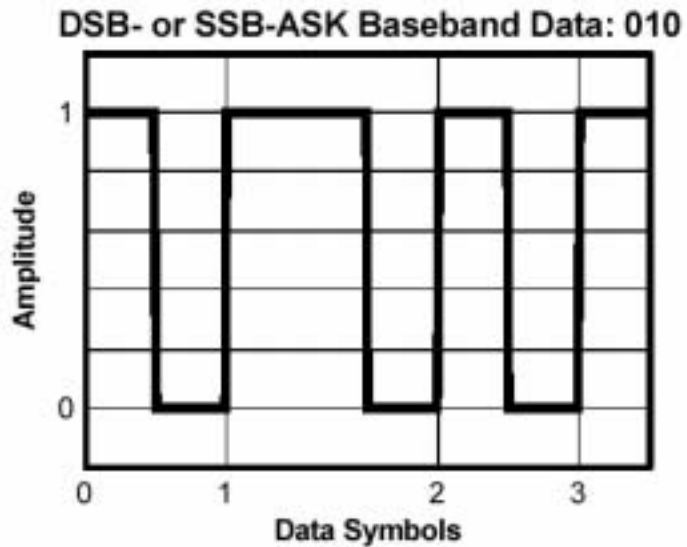


SSB-ASK (2)

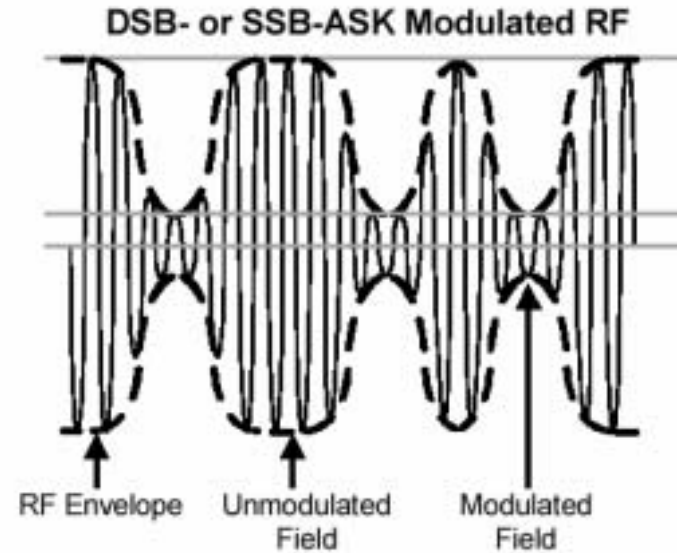


DSB-ASK and SSB-ASK

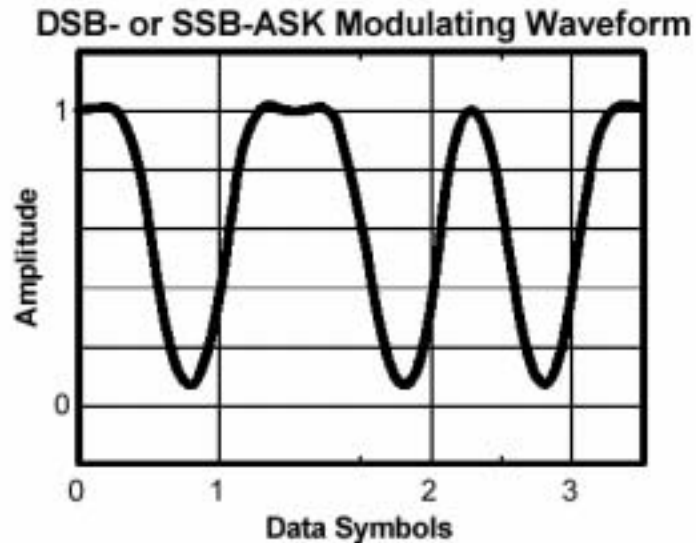
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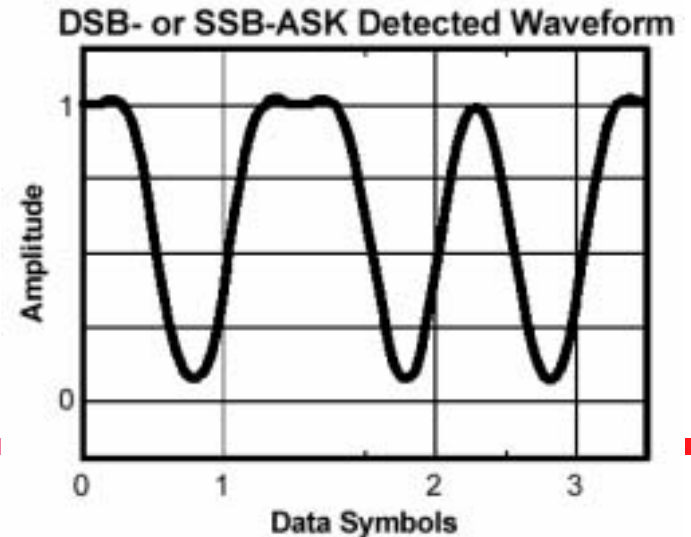
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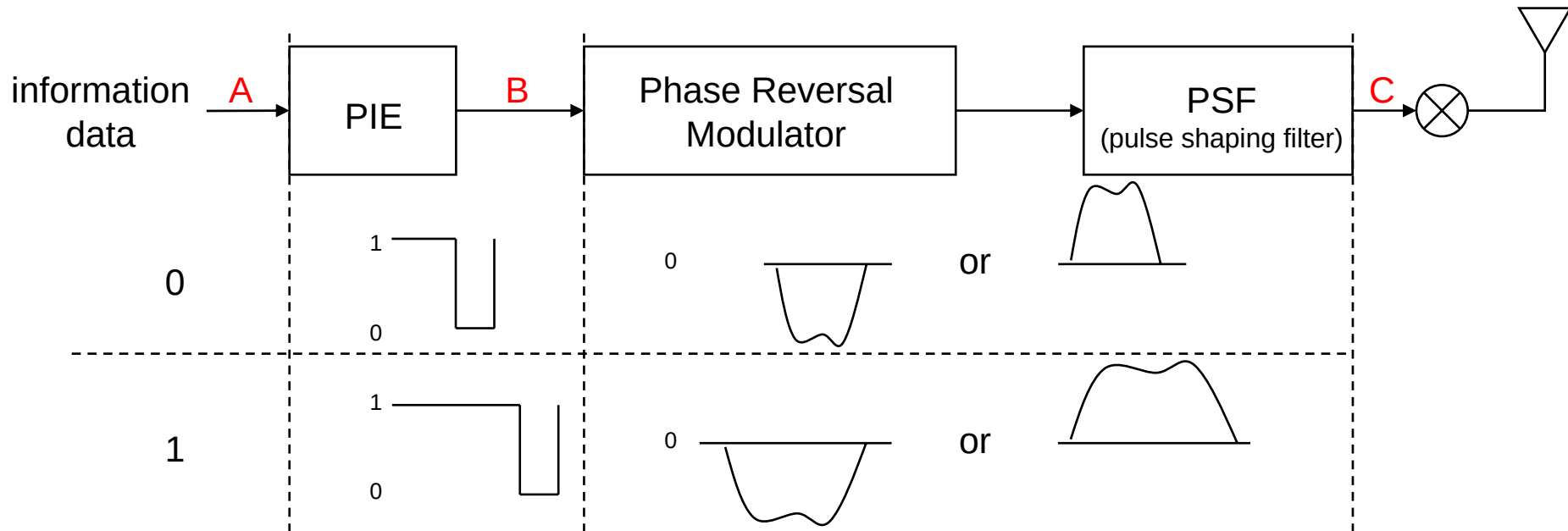
2



4



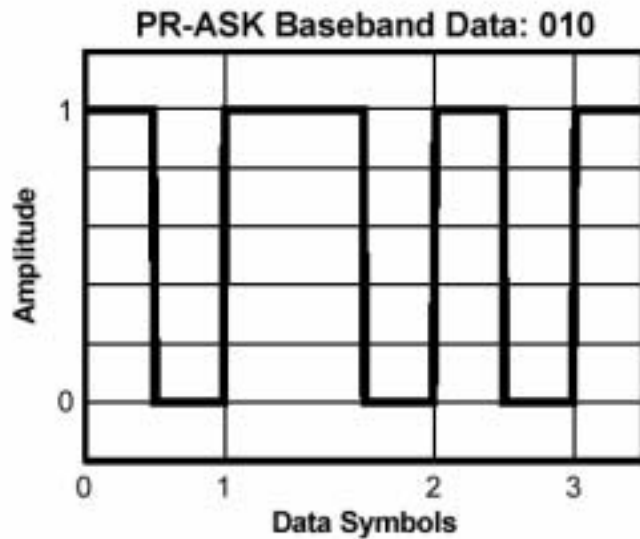
PR-ASK (1)



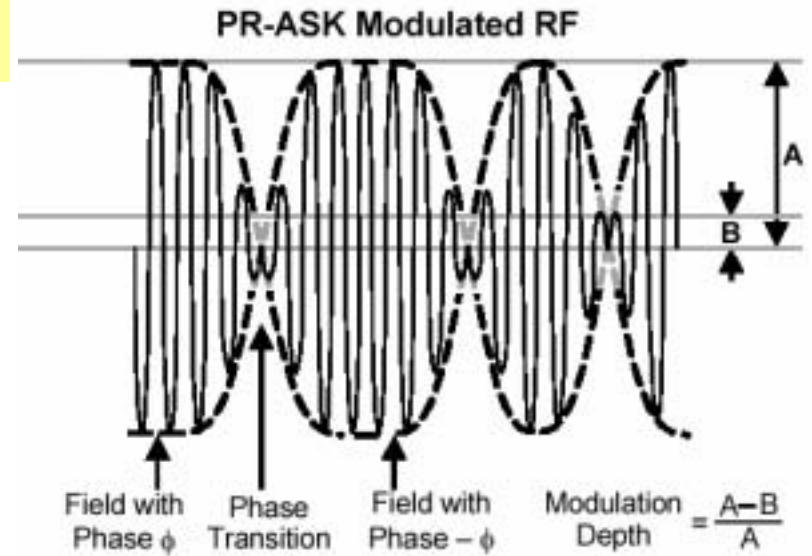
- Bandwidth of PR-ASK signal is more narrow than that of DSB-ASK signal since the sign of modulating waveform does not change during the symbol period.
 - Modulating waveform for data '0' and '1' has zero-power interval.

PR-ASK (2)

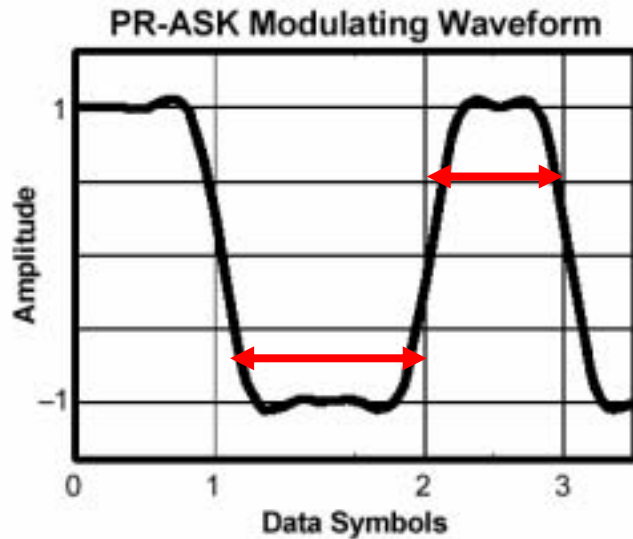
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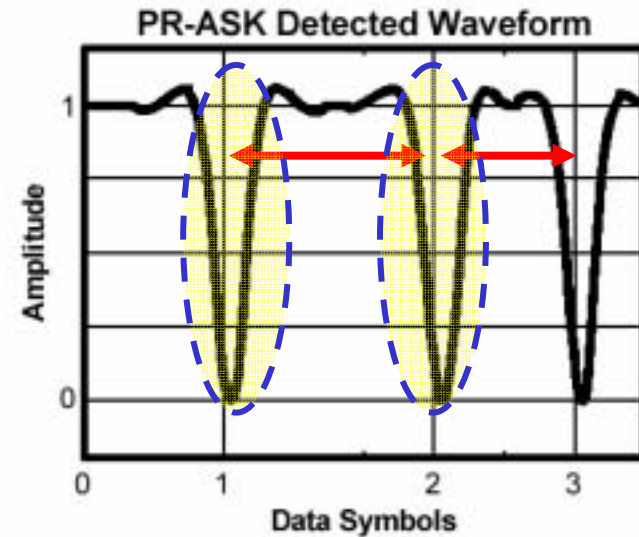
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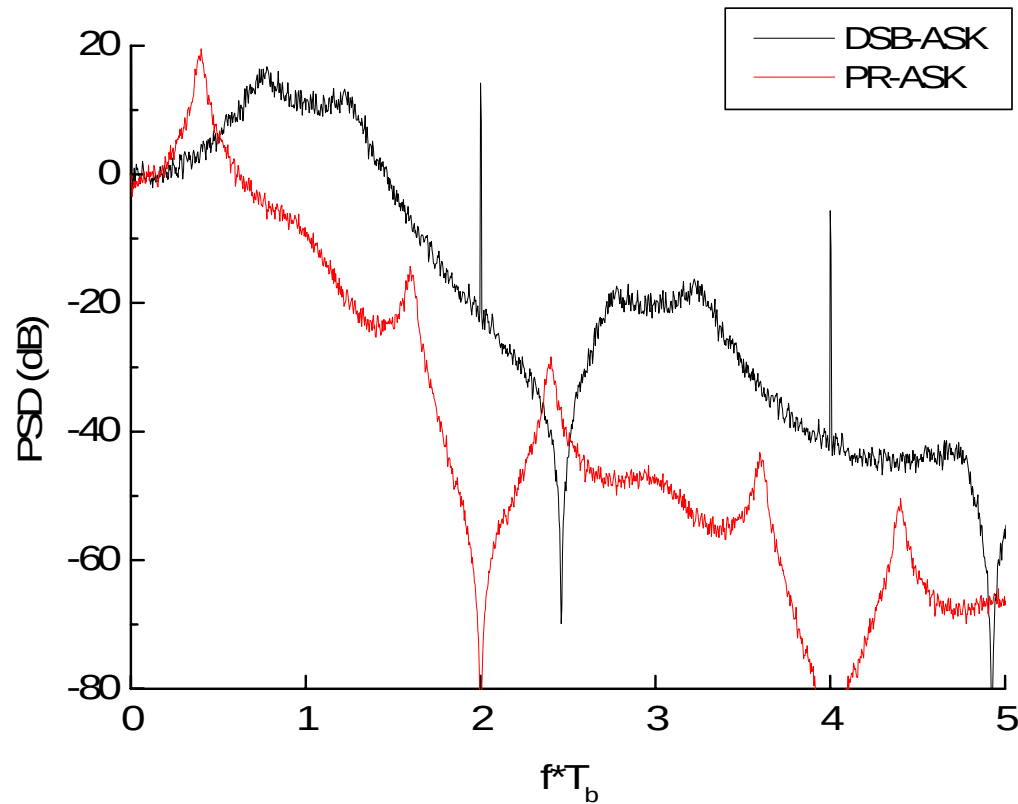
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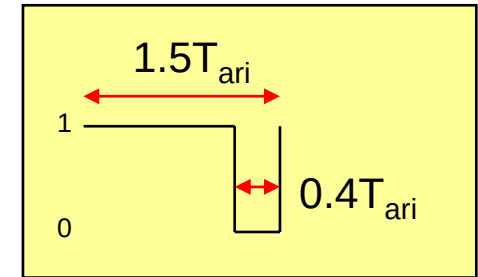
4



Comparison of power spectrums



PIE



	99% energy bandwidth
DSB-ASK	$3.5/T_{ari}$
SSB-ASK	$1.75/T_{ari}$
PR-ASK	$1.62/T_{ari}$

EPCglobal RFID protocol:
Tag→Interrogator(Reader) reception

Features

- Modulation
 - ASK
 - PSK
- Subcarrier frequency
 - 40kHz ~ 640kHz
- Subcarrier modulation
 - Miller modulation.
- Data coding
 - FM0 baseband.
 - Miller modulation.
- Data rates
 - FM0 : 40kbps ~ 640kbps
 - Miller: 5kbps(M=8)~320kbps(M=2)

FM0 baseband (1)

- It specifies the phase inversion on a symbol boundary.

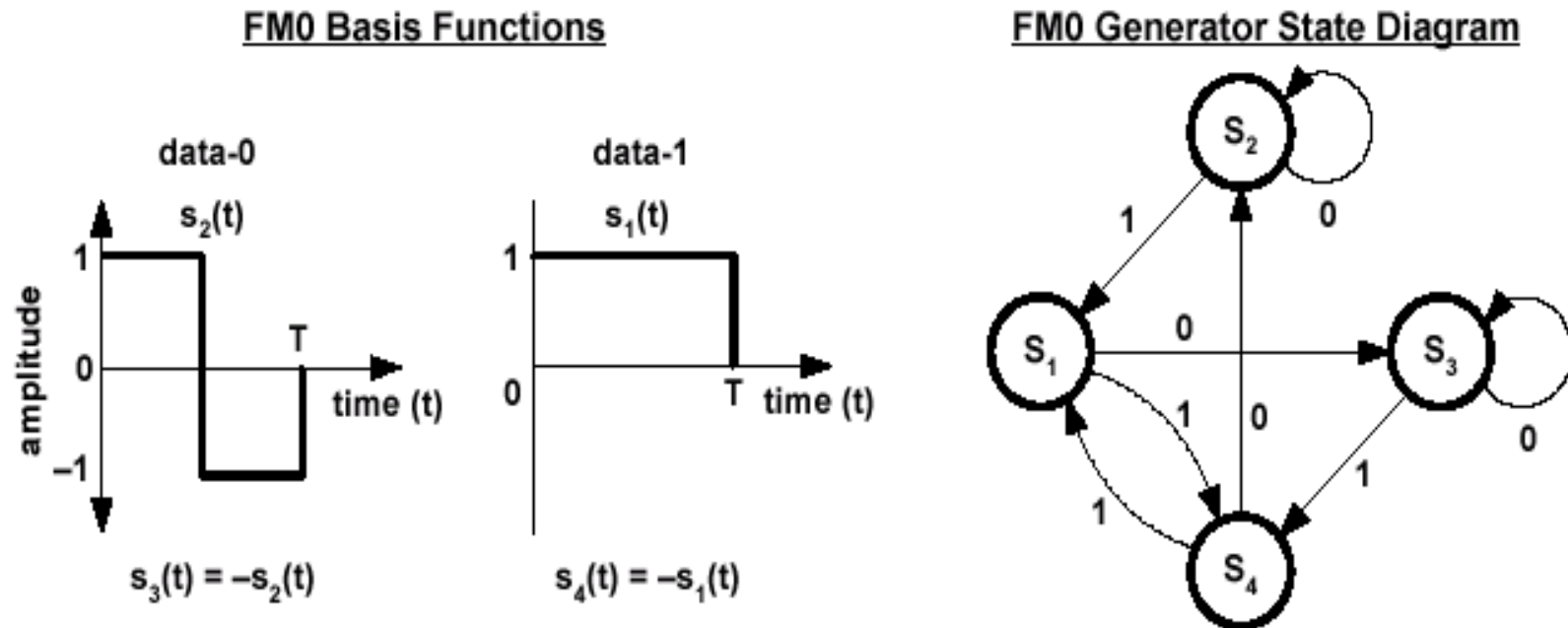


Figure 6.8 – FM0 basis functions and generator state diagram

FM0 baseband (2)

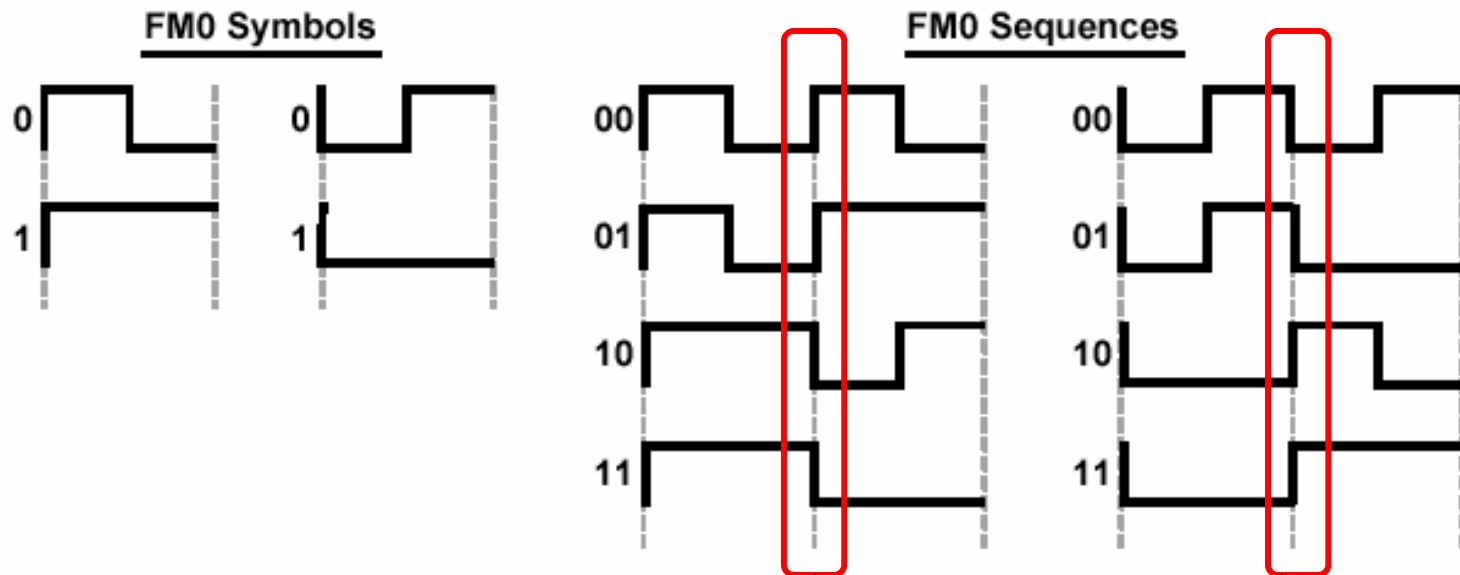
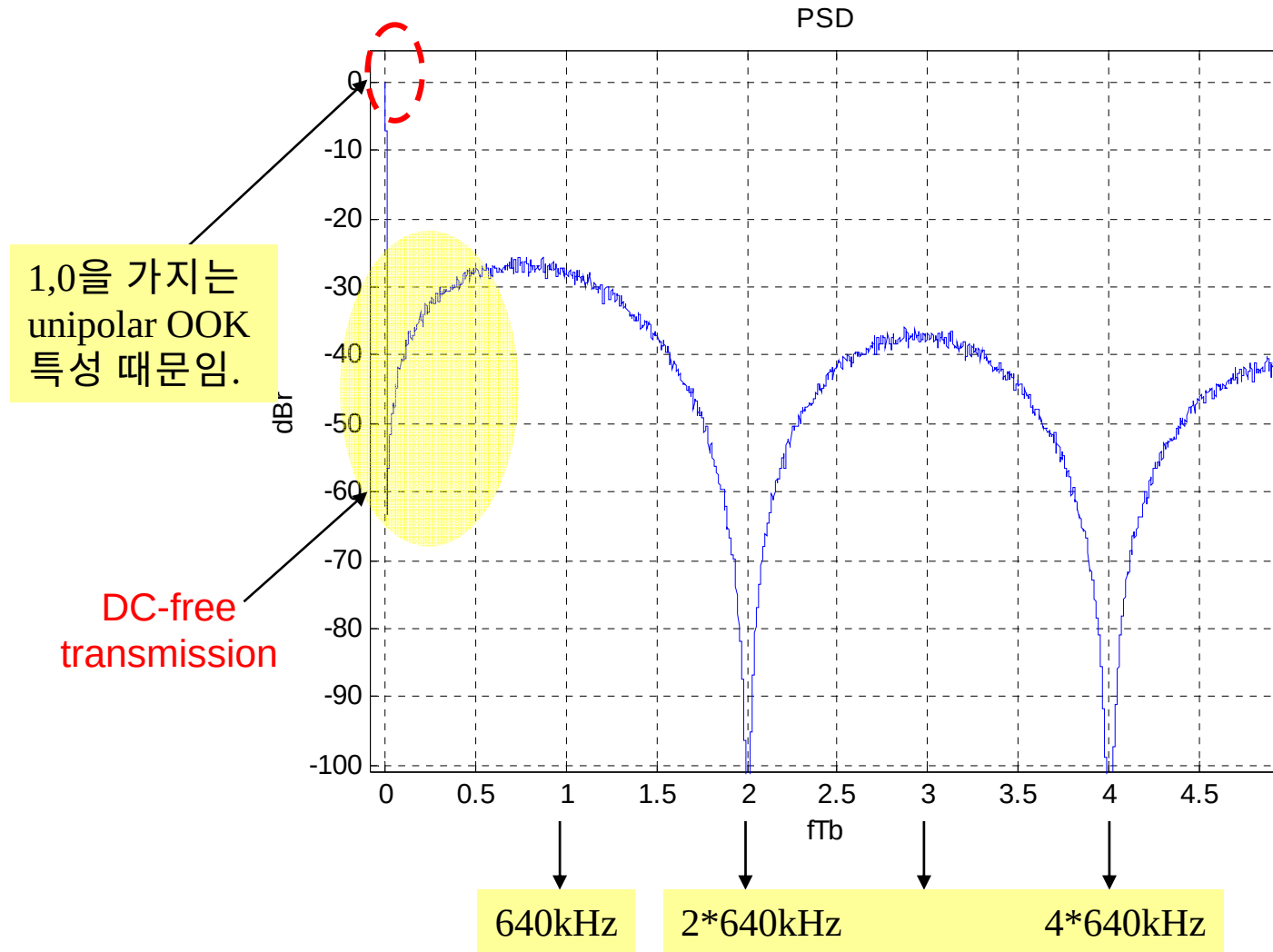


Figure 6.9 – FM0 symbols and sequences

Phase inversion

PSD of FM0-OOK (LF=640kHz)



Miller modulation (1)

- The transmitted waveform is the baseband waveform multiplied by a square wave at M times the symbol rate.
 - subcarrier cycles per bit (M) : 2, 4, 8.
- It specifies the phase keep on a symbol boundary.
 - Phase on a symbol boundary is inverted by using a subcarrier.

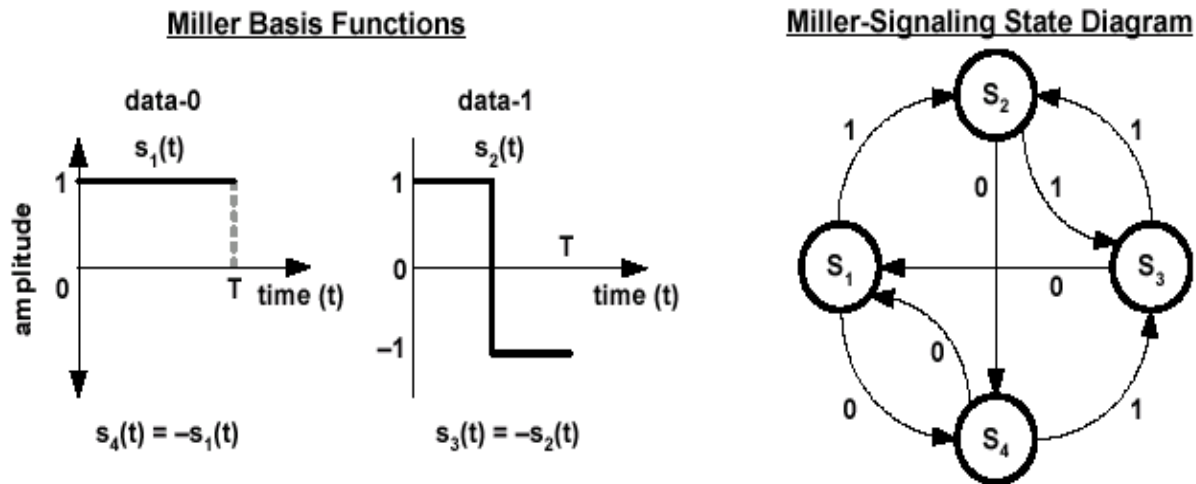


Figure 6.12 – Miller basis functions and generator state diagram

Miller modulation (2)

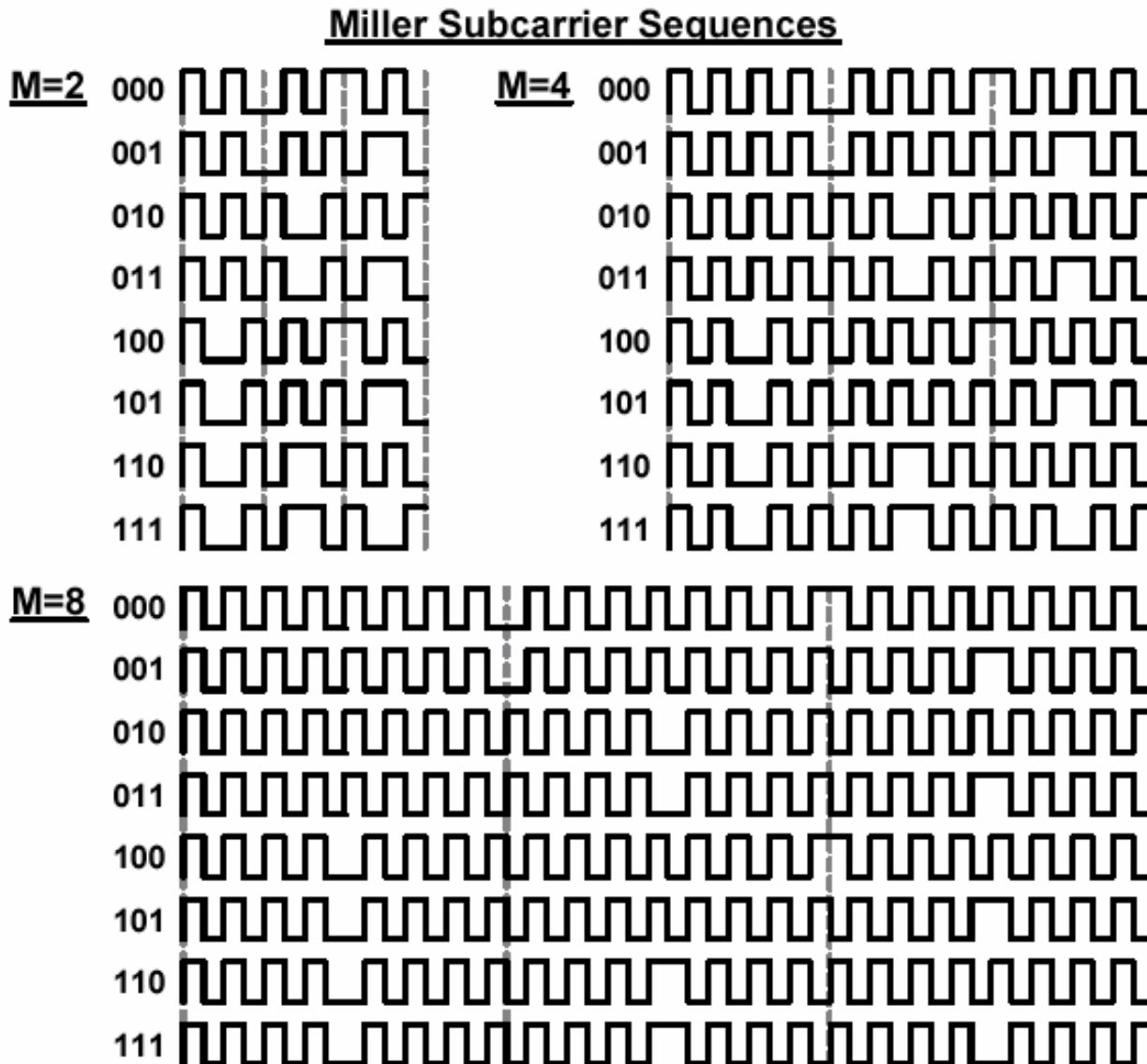
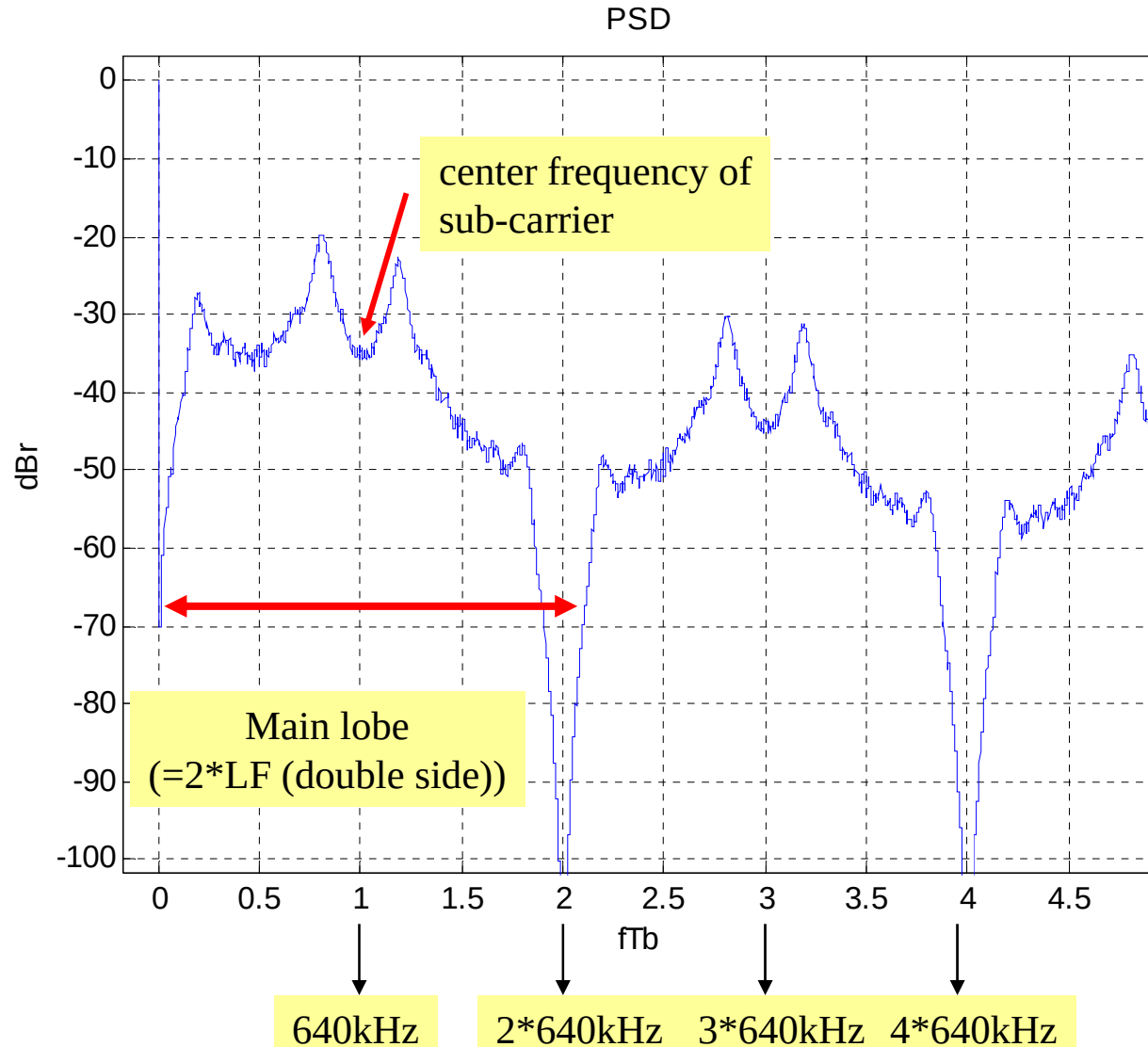


Figure 6.13 – Subcarrier sequences

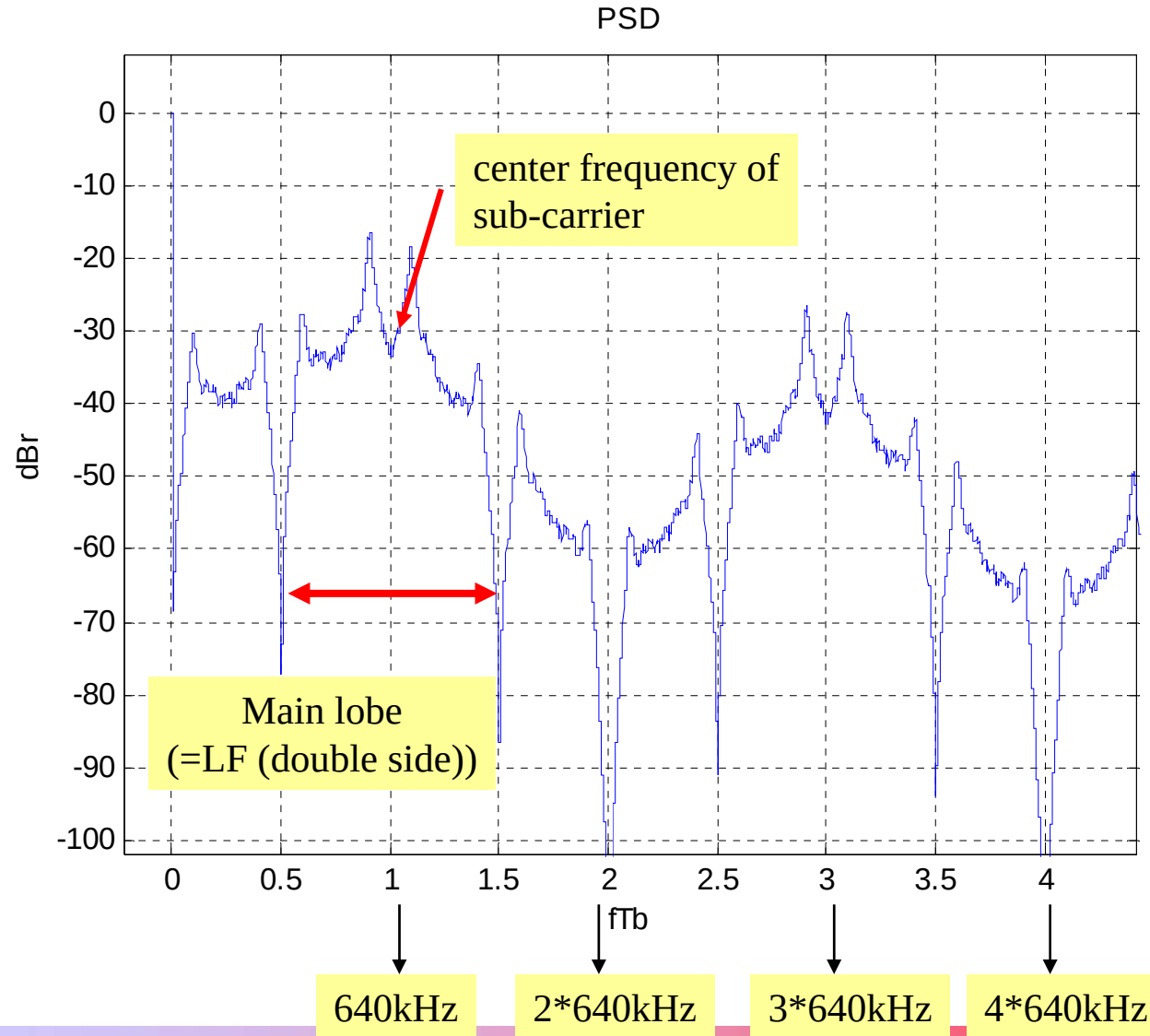
Tag-to-Interrogator data rates

M: Number of subcarrier cycles per symbol	Modulation type	Data rate (kbps)
1	FM0 baseband	LF
2	Miller subcarrier	LF/2
4	Miller subcarrier	LF/4
8	Miller subcarrier	LF/8

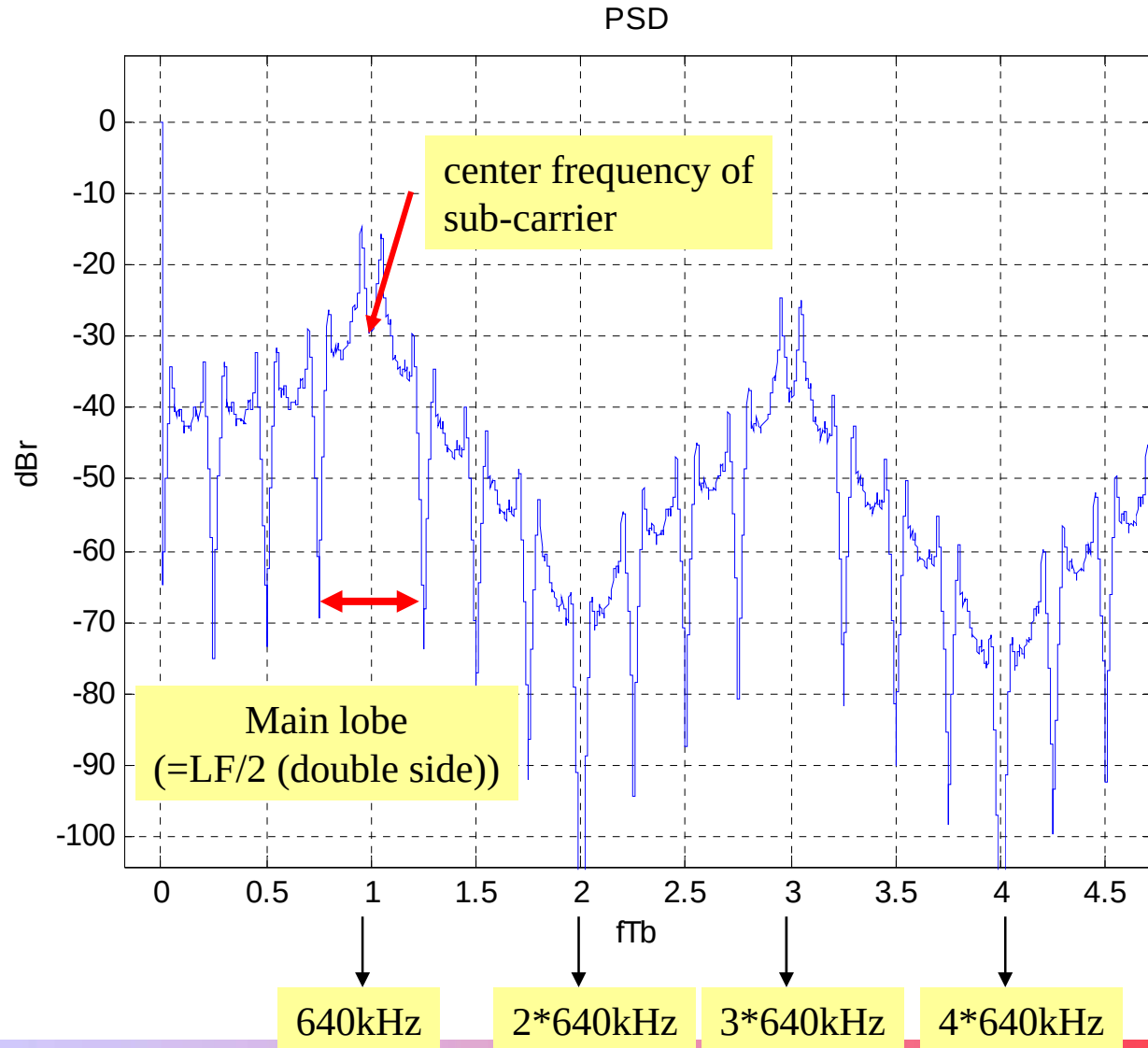
PSD of Miller-OOK (LF=640kHz, M=2)



PSD of Miller-OOK (LF=640kHz, M=4)



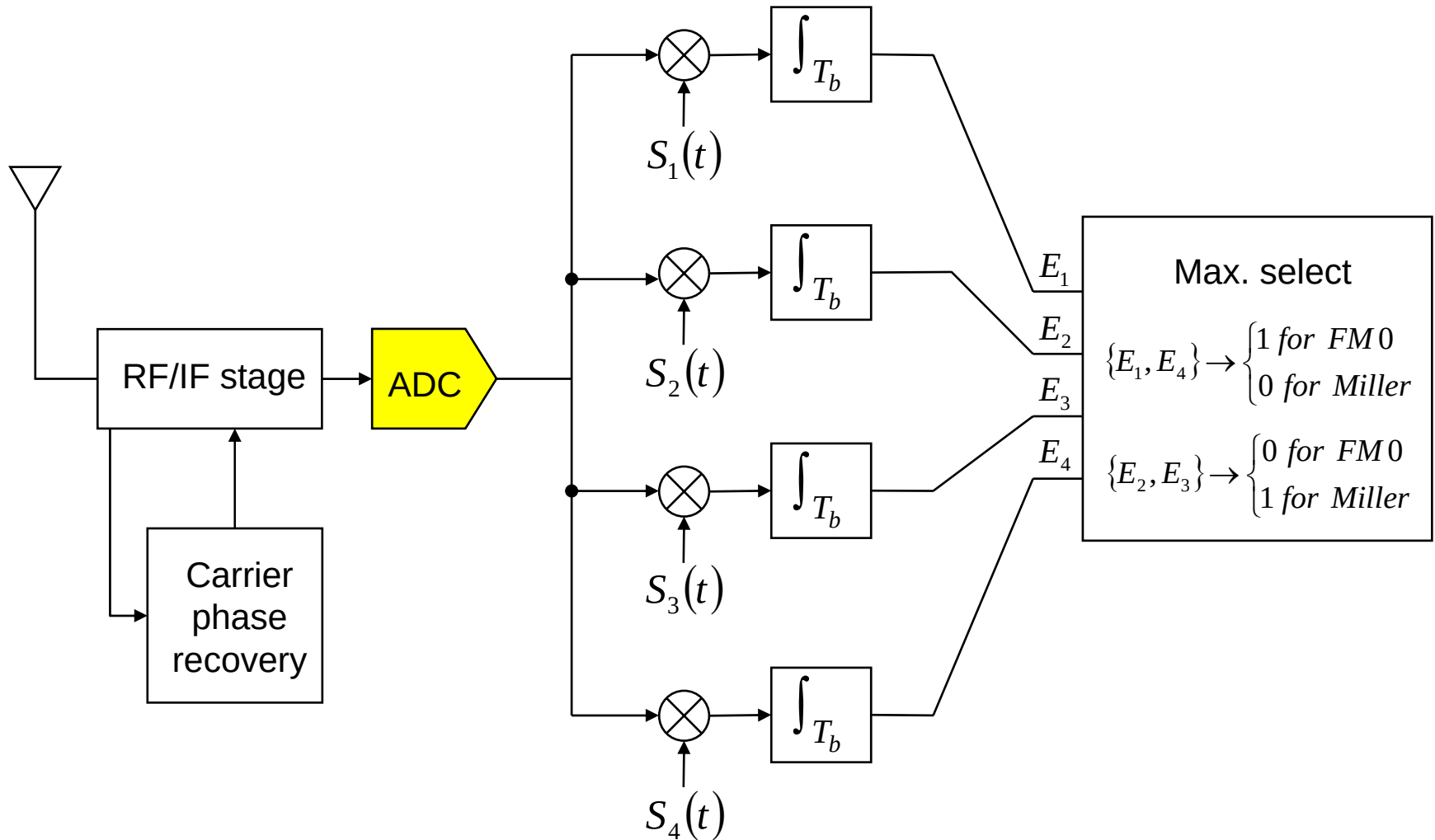
PSD of Miller-OOK (LF=640kHz, M=8)



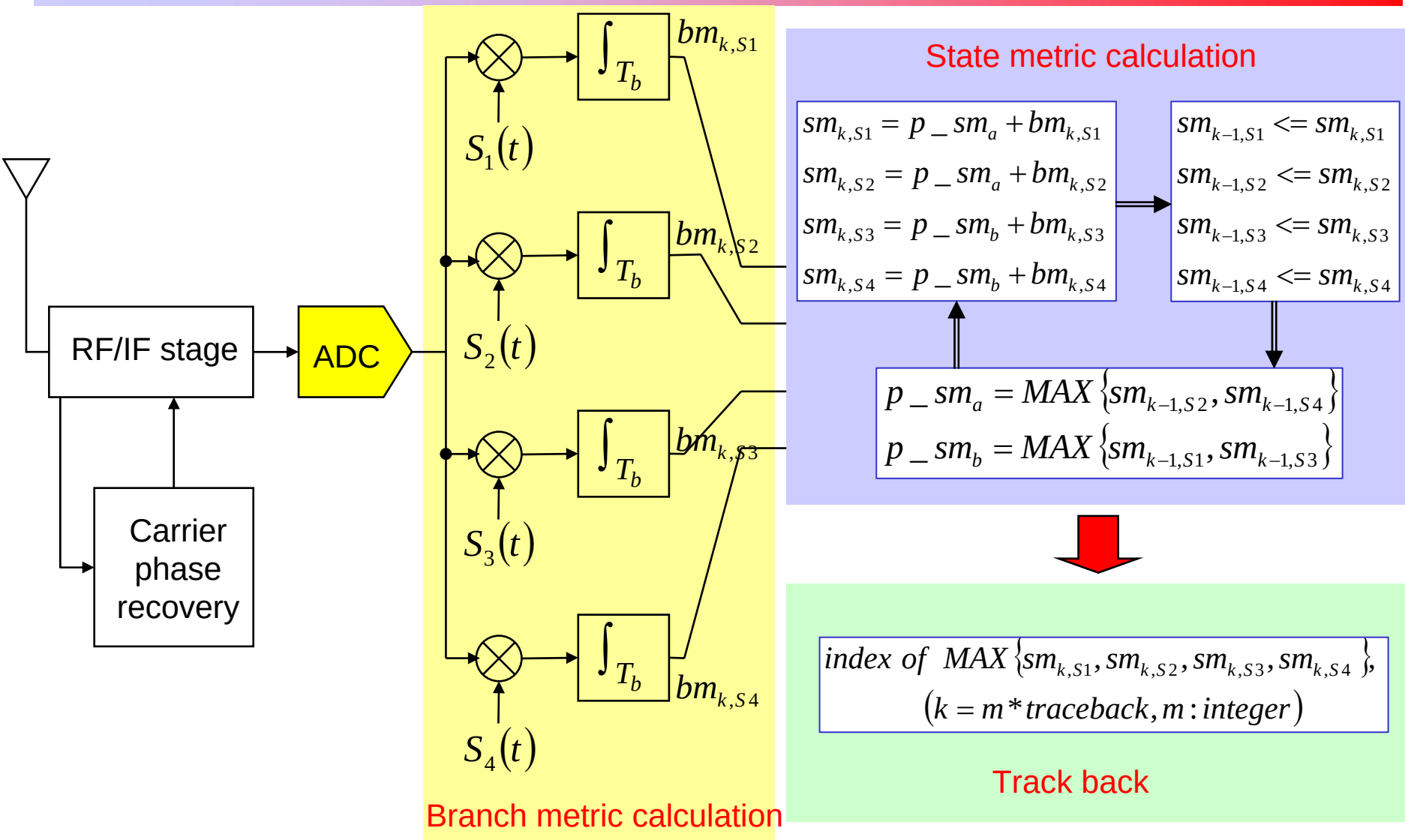
Demodulator for RFID reader

- Coherent detection
 - The coherent demodulator for RFID reader can be simpler than conventional coherent demodulator since the received carrier of the RFID reader is identical to the local carrier except carrier phase.
 - Allowed types
 - Symbol-by-symbol detection
 - Trellis detection
- Non-coherent detection
 - FM0 or Miller encoded ASK(OOK)/BPSK signal is equal to differential BPSK (DBPSK).
 - It allows to use differential detection for RFID reader.

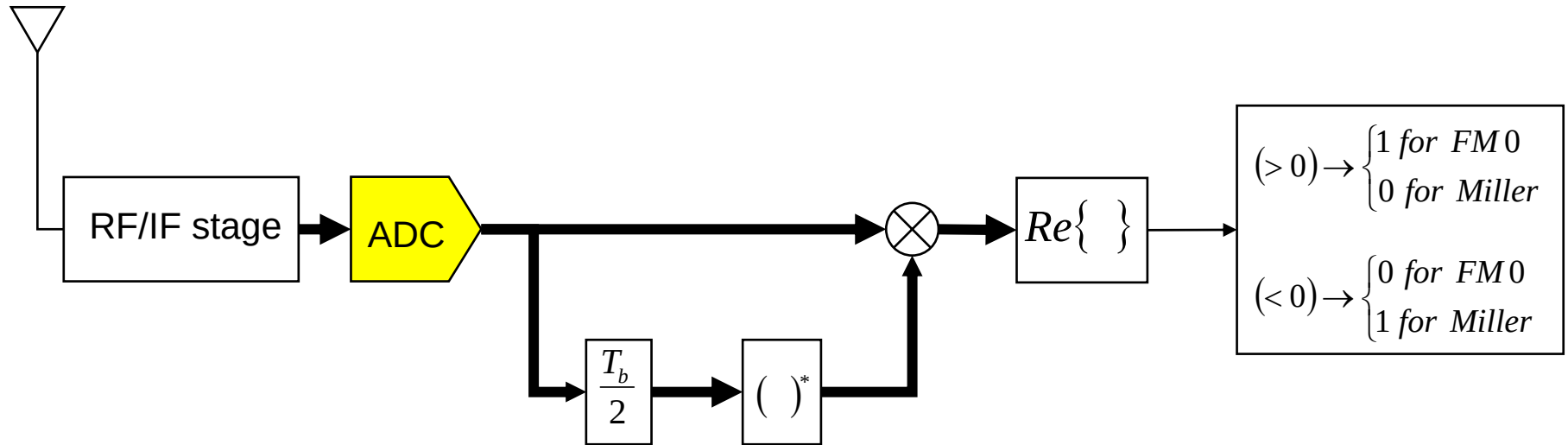
Block diagram of symbol-by-symbol coherent detection



Block diagram of coherent detection with trellis decoding



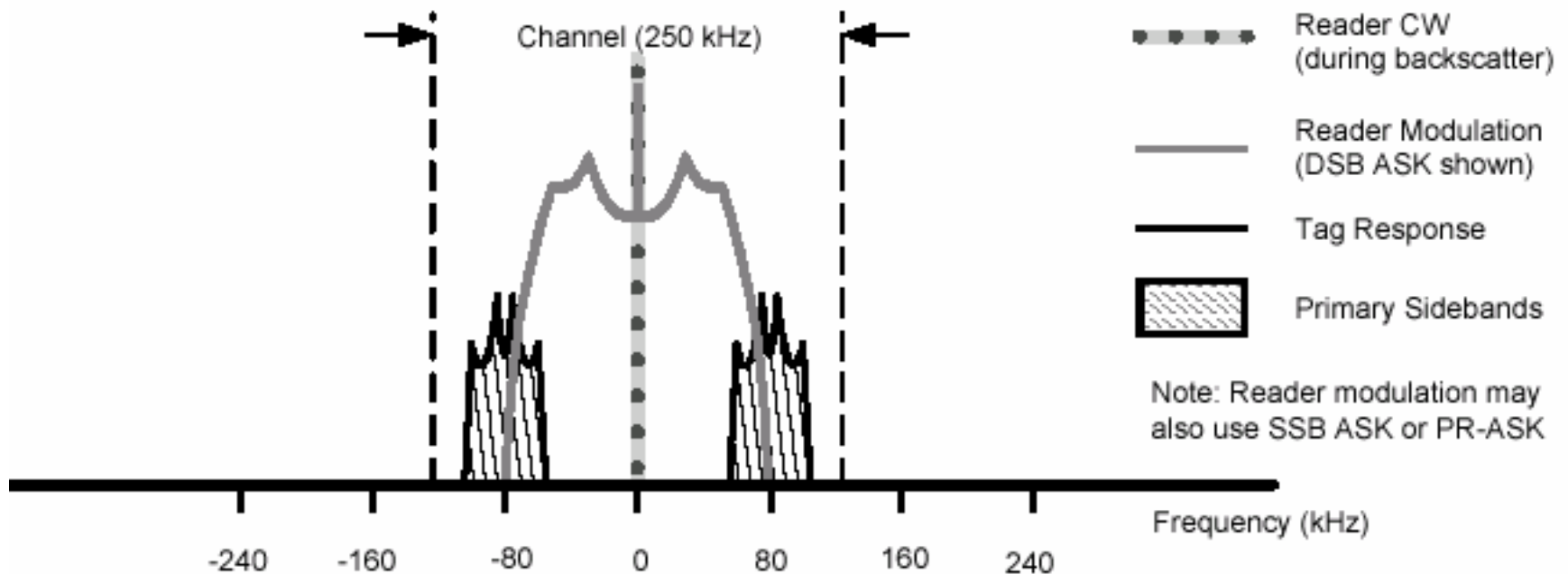
Block diagram of differential detection based non-coherent detection



Dense-interrogator channelized signaling (1)

- Single-channel regulatory environment (CEPT)
 - Interrogator spectrum and tag spectrum in the same channel.
 - Interrogator transmission and tag responses are separated temporally.

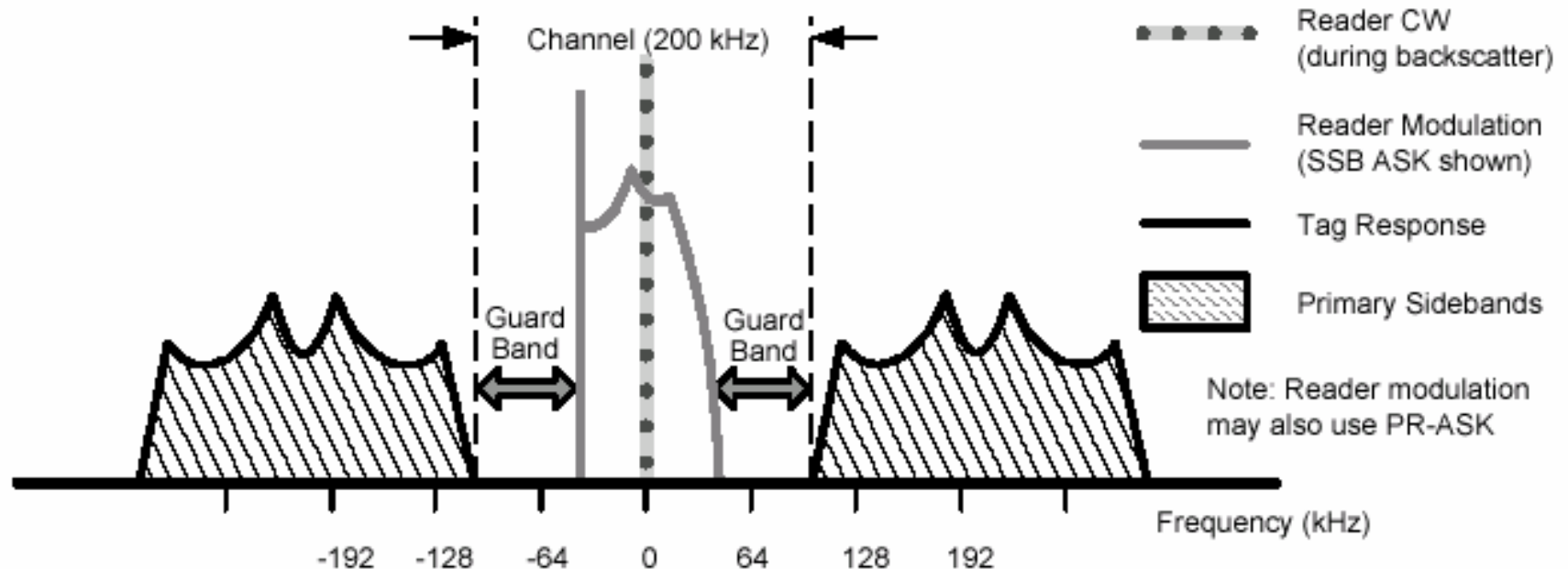
A: Subcarrier Spectral Allocation (CEPT: Single Channel)



Dense-interrogator channelized signaling (2)

- Multi-channel regulatory environment (CEPT)
 - Interrogator transmission and tag responses are separated spectrally.
 - Interrogator transmissions are located in even-numbered channels.
 - Tag backscatter is located in odd-numbered channels with miller-encoding.

B: Subcarrier Spectral Allocation (CEPT: Multiple Channels)



Dense-interrogator channelized signaling (3)

- FCC environments
 - Interrogator transmission and tag responses are separated spectrally.
 - Interrogator transmissions is centered in channels.
 - Tag backscatter is located at channel boundaries.

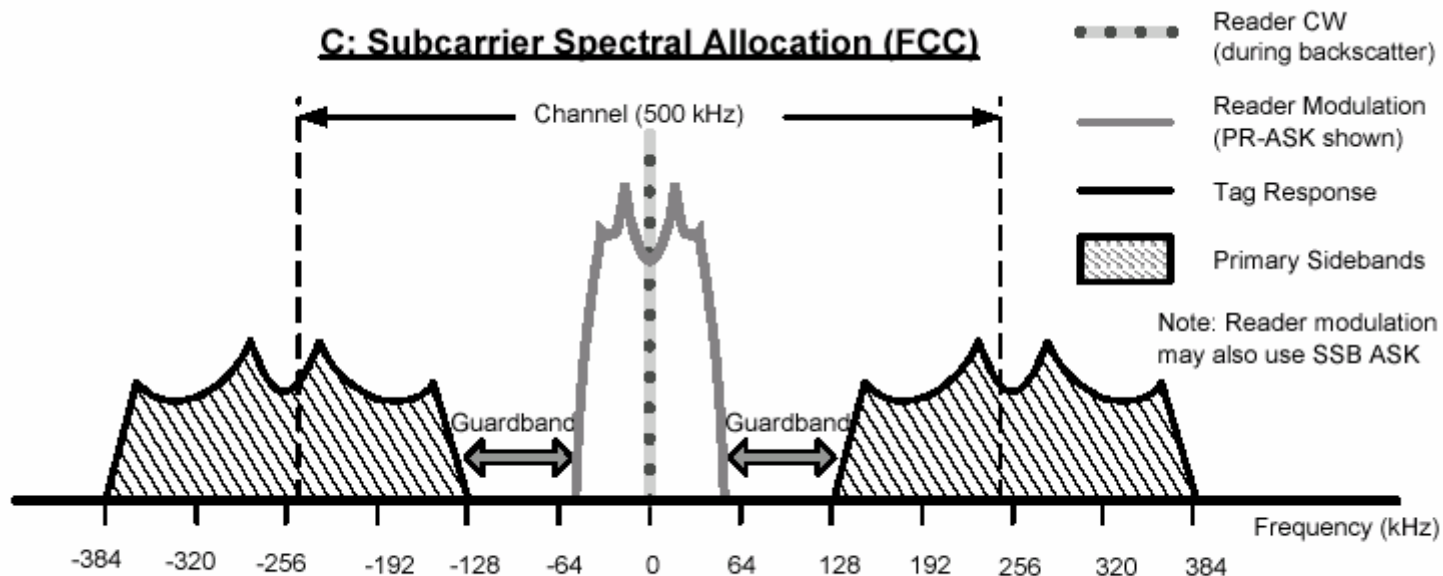


Figure G.2 – Subcarrier spectral allocation for CEPT and FCC environments

Dense-interrogator channelized signaling (4)

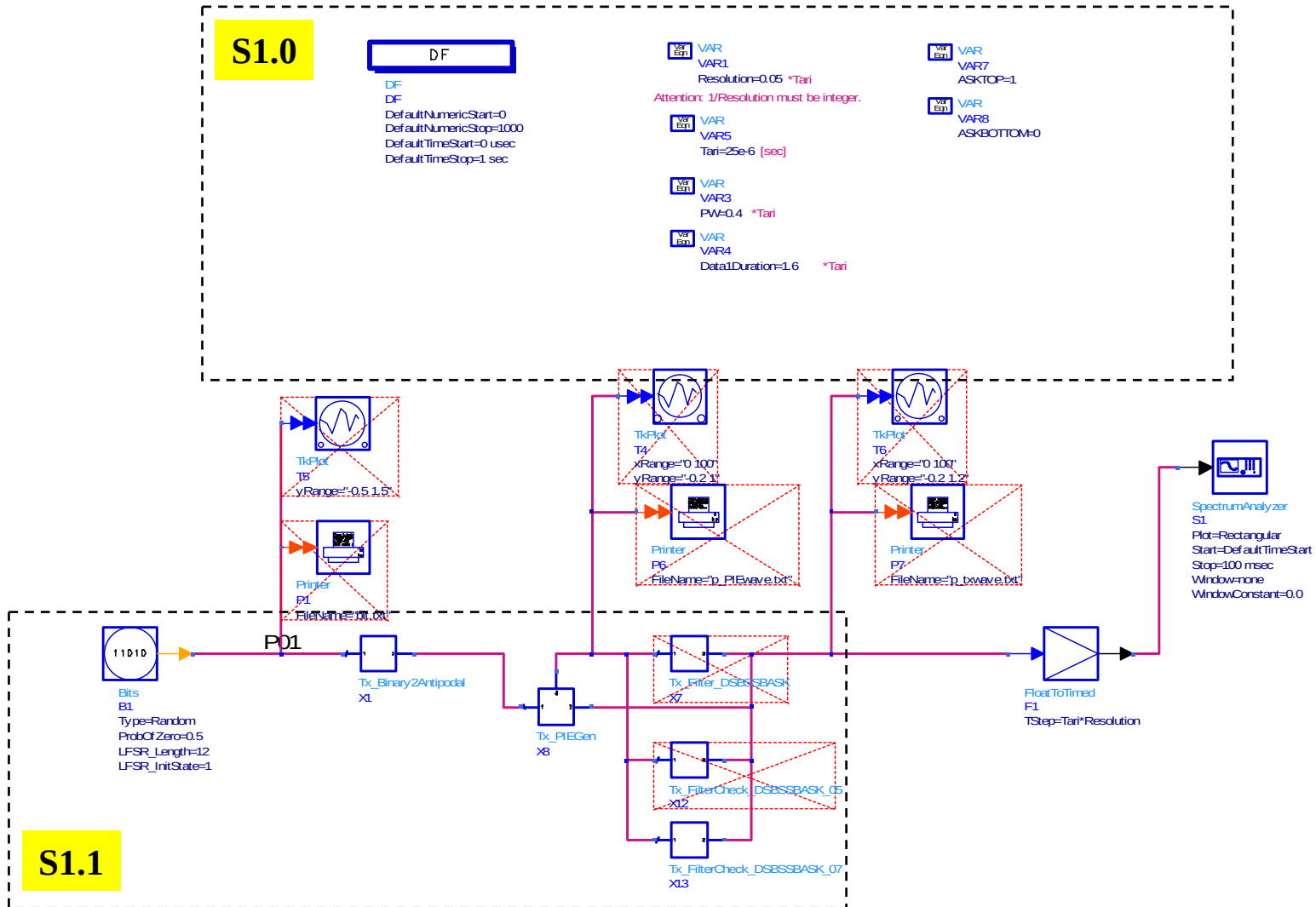
	Single-channel	Multi-channel	FCC
Characteristics	1. Wide channel. 2. Fewer channels	1. Narrow channel. 2. More channels	-
Reader	DSB-ASK (wider-bandwidth)	SSB-ASK	-
Tag	FM0 , Miller (M=2) (small frequency offset)	Miller (M=4,8) (large frequency offset)	Miller (M=2,4) (controlled frequency offset)

Conclusions

- RFID has been introduced.
- UHF RFID MODEM (standardized by EPCglobal) has been reviewed.
 - Interrogator→Tag transmission.
 - PIE
 - DSB/SSB/PR-ASK
 - Tag→Interrogator reception.
 - FM0, Miller subcarrier baseband encoding
 - Reader demodulators
 - Coherent detection: symbol-by-symbol detection, trellis detection
 - Differential detection based non-coherent detection
 - Dense-interrogator channelized signaling and Modems

UHF RFID (900MHz, EPCglobal Gen.2) - Agilent ADS System Simulator

S1. Reader-to-Tag transceiver - block diagram



S1.0 Parameters

DF

1bit를 표현하는
sample의 개수의 역수

DF
DF
DefaultNumericStart=0
DefaultNumericStop=100
DefaultTimeStart=0 usec
DefaultTimeStop=1 sec

Tari 값을
setting한다.

1bit 전송의 끝을 알리는
low pulse의 시간을 setting
한다 (Tari에 대한 비율).

data-1 pulse의 시간
(Tari에 대한 비율)

Var
Eqn

VAR

VAR1

Resolution=0.05 *Tari

Attention: 1/Resolution must be integer.

Var
Eqn

VAR

VAR5

Tari=25e-6 [sec]

Var
Eqn

VAR

VAR3

PW=0.4 *Tari

Var
Eqn

VAR

VAR4

Data1Duration=1.6 *Tari

Var
Eqn

VAR

VAR7

ASKTOP=1

ASK modulated
waveform에서
최대값

Var
Eqn

VAR

VAR8

ASKBOTTOM=0

ASK modulated
waveform에서
최소값

Data1Duration (*Tari)

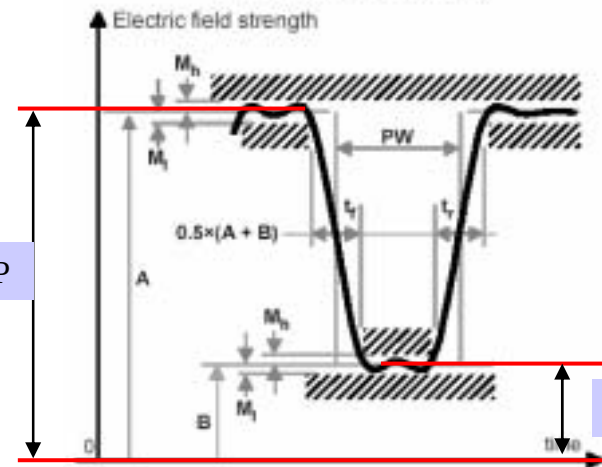
Tari

PW (*Tari)

data-0

data-1

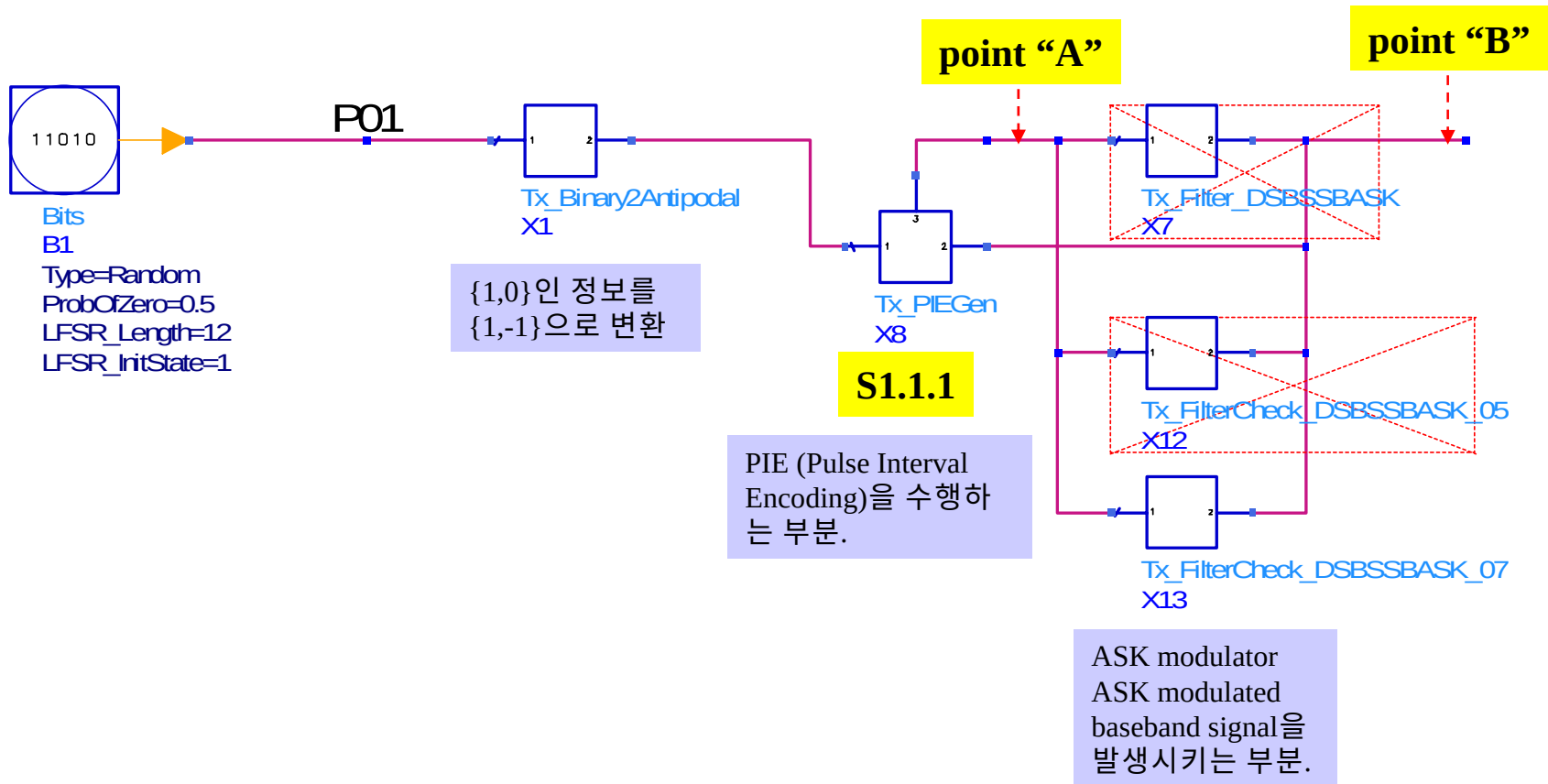
ASK Modulation



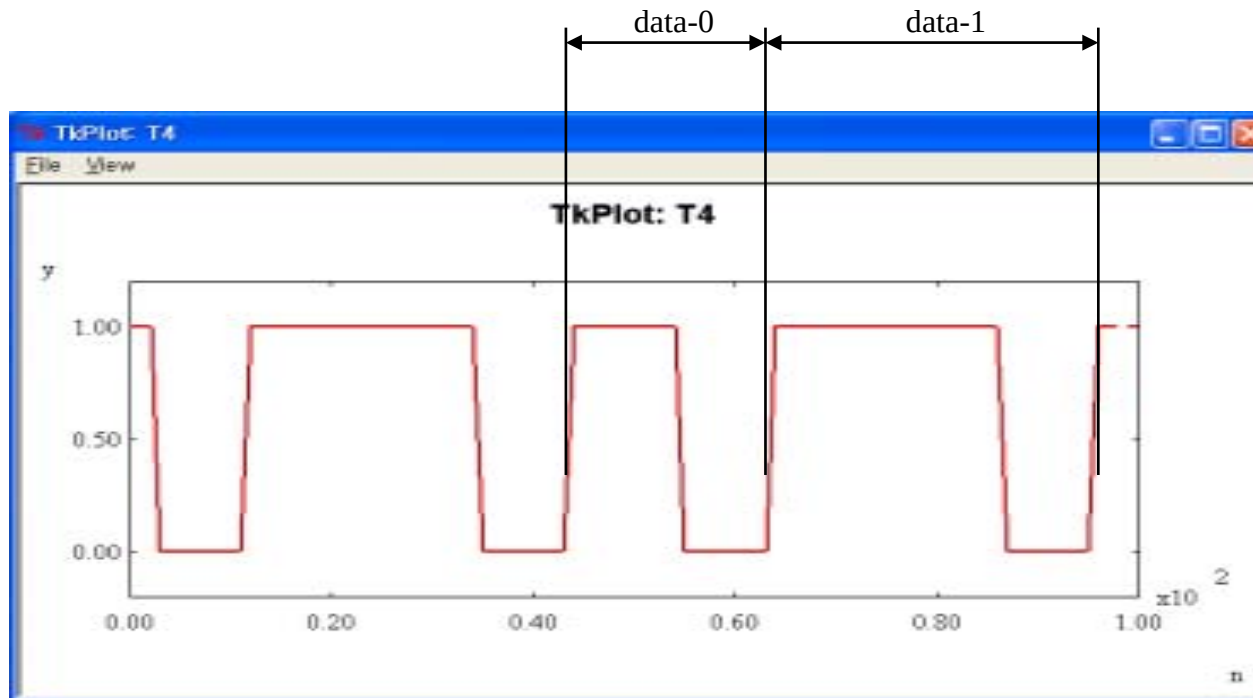
ASKTOP

ASKBOTTOM

S1.1 Transmitter

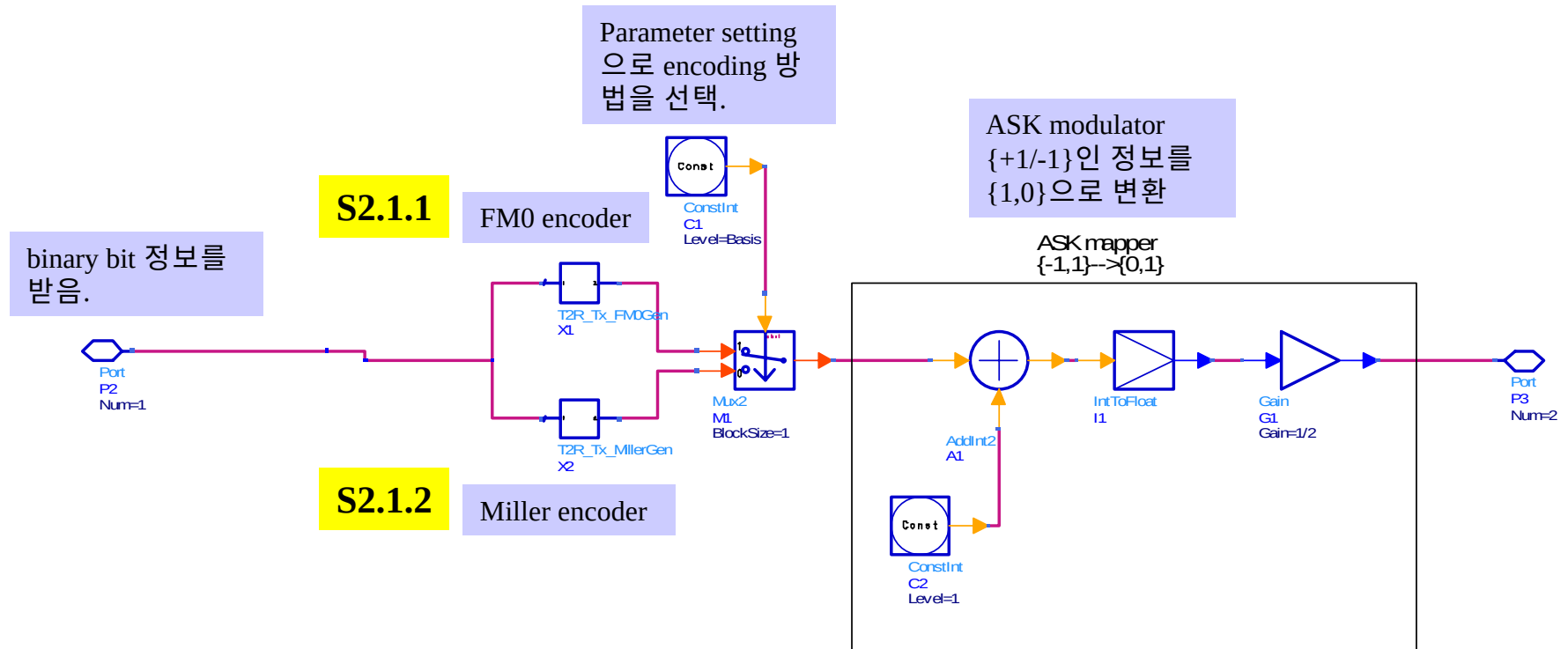


S1.1 point “A” 결과 plot - PIE encoded waveform

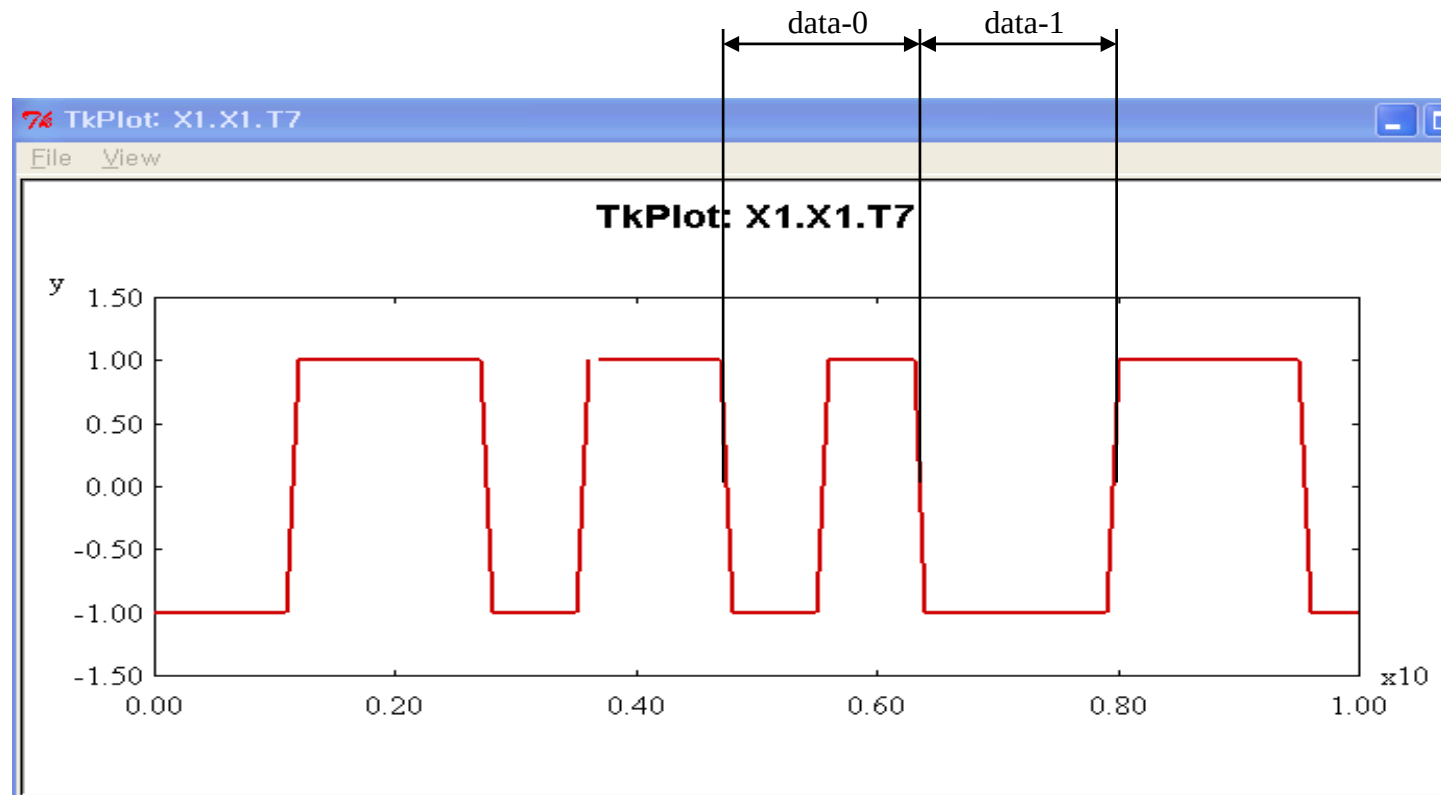




S2.1 Transmitter

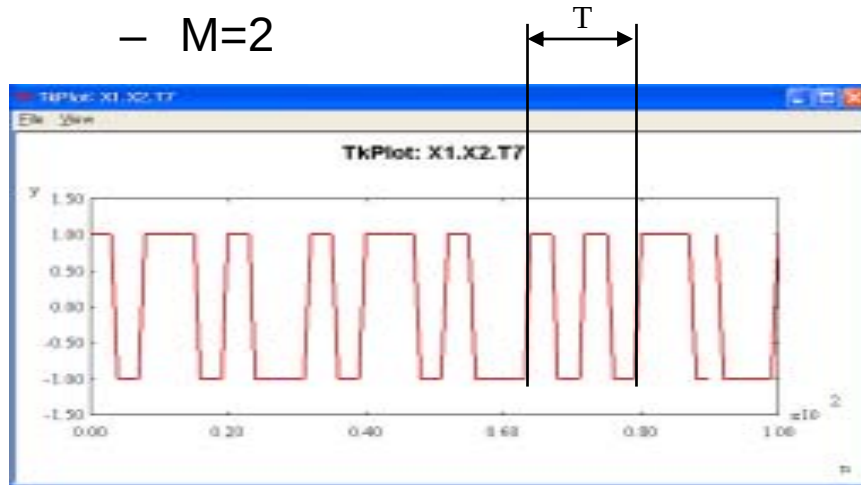


S2.1.1 FM0 encoded waveform

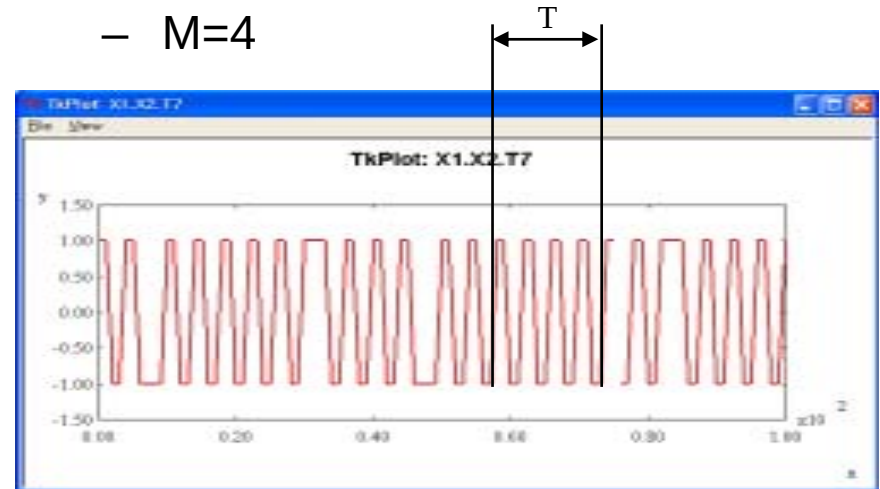


S2.1.2 Miller encoded waveforms

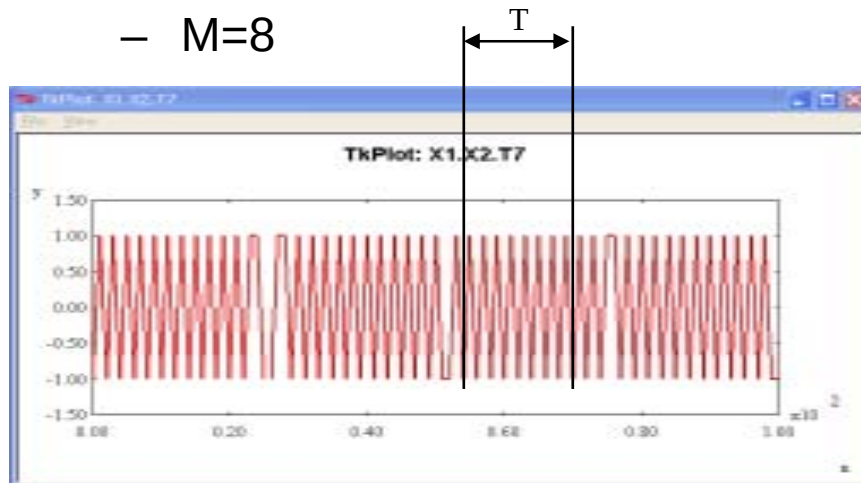
– M=2



– M=4

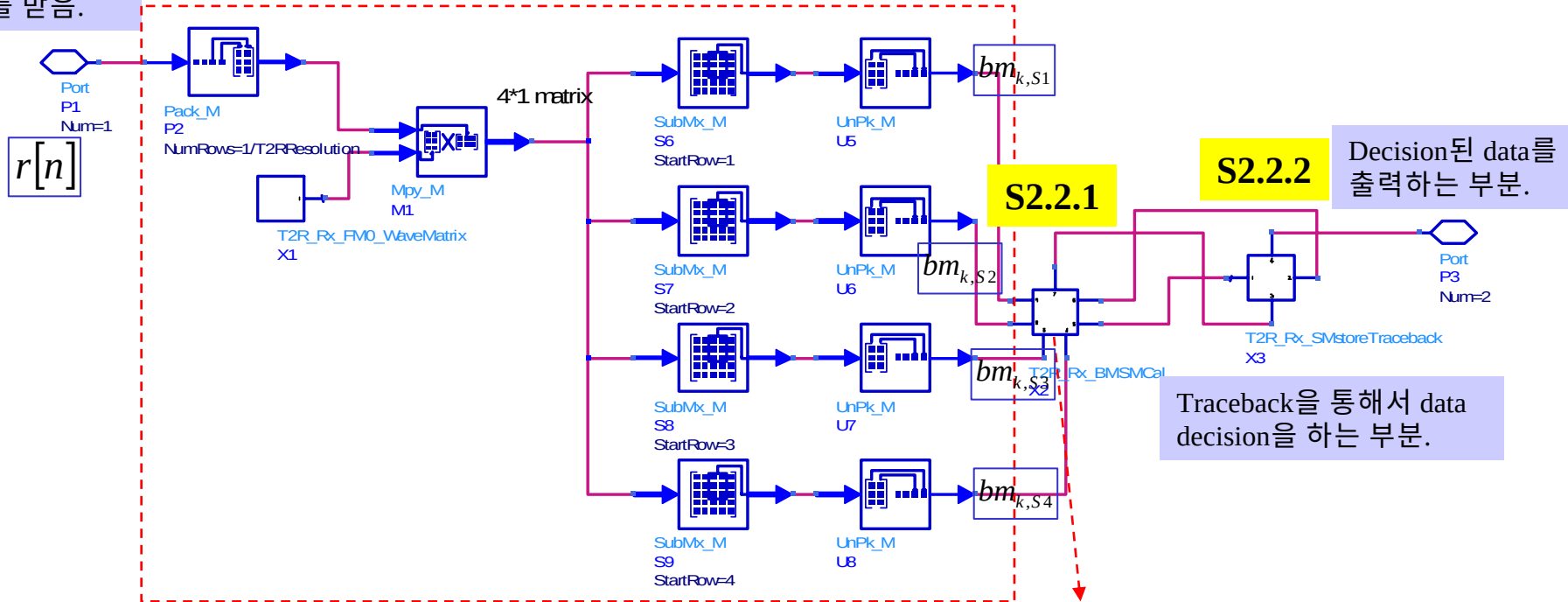


– M=8



S2.2 Receiver

수신 신호
를 받음.



1T동안의 발생가능한 모든
waveform과 수신신호의
distance를 계산하는 부분.
(FM0 encoding 지원)

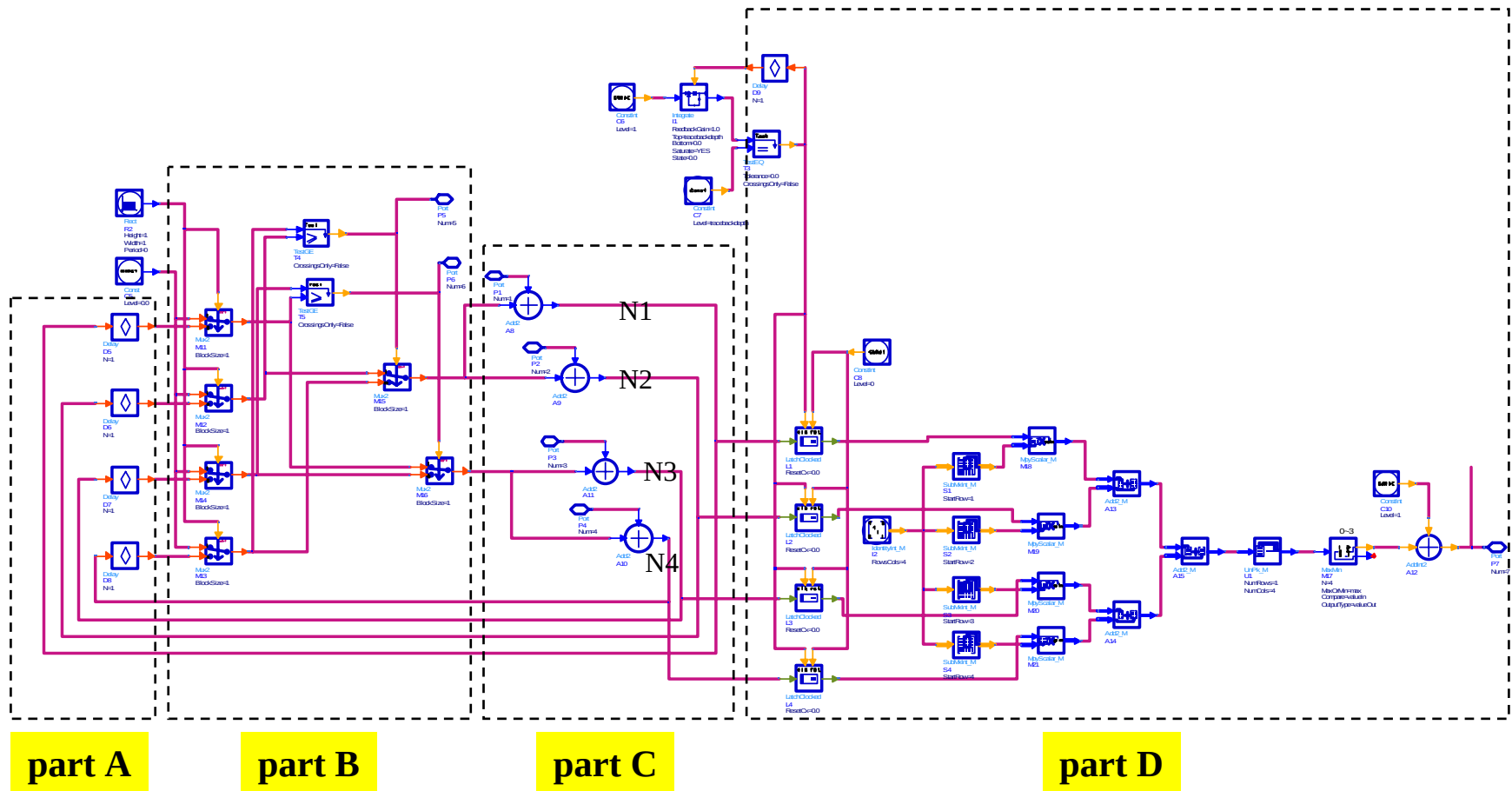
Trellis decoding을 위한
branch metric과 state
metric 계산 부분.

Traceback을 통해서 data
decision을 하는 부분.

Decision된 data를
출력하는 부분.

$$bm_{k,Si} = \sum_{n=0}^{N-1} (r[k \cdot N + n] \cdot s_i[n]), \quad (i = 1, 2, 3, 4)$$

S2.2.1 Trellis decoding - metric calculation



S2.2.2 Trellis decoding - traceback

